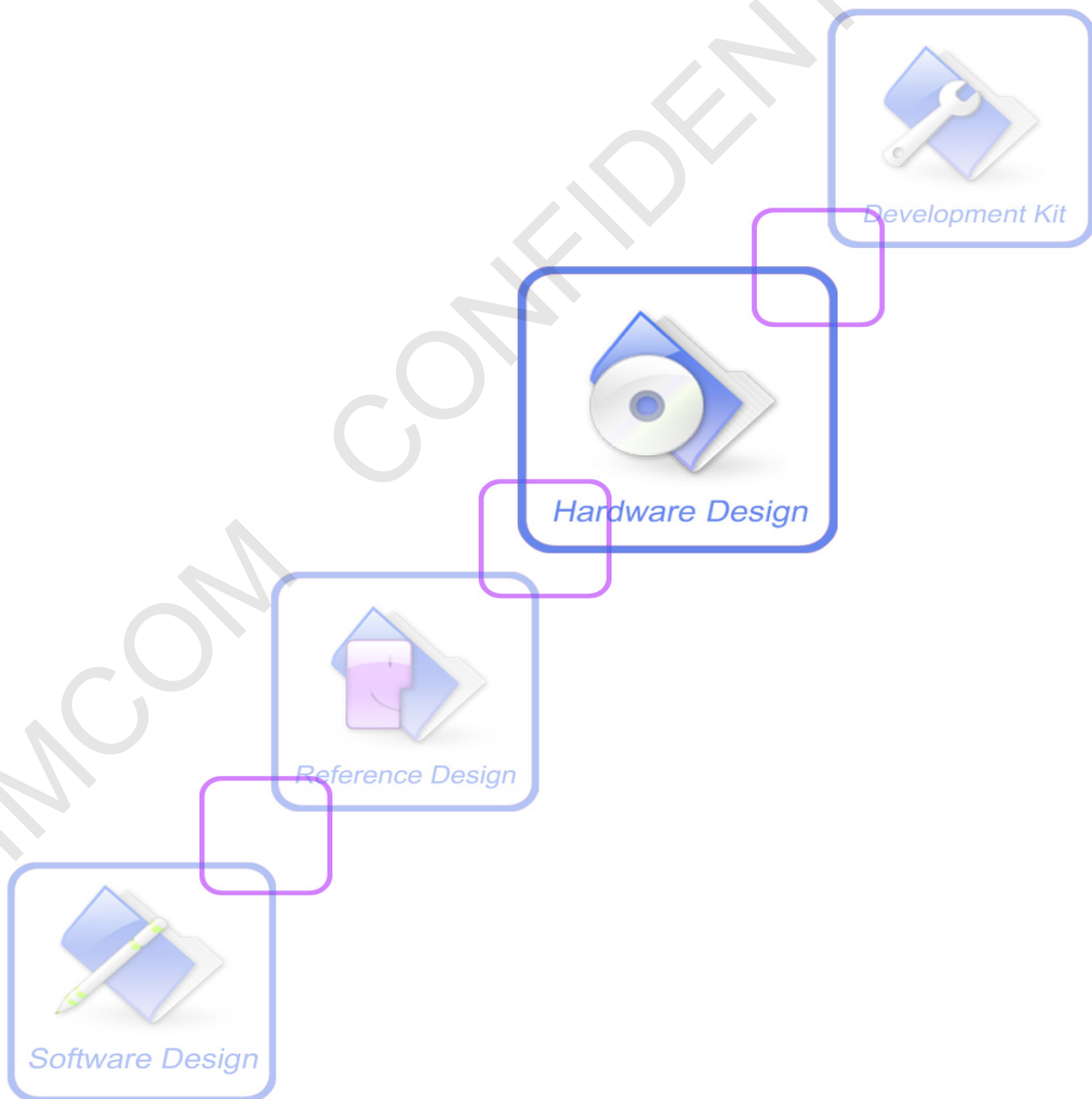


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2015-04-27	1.02	1. Update figure7,figure10,figure11,figure10,figure12,figure13,figure14,figure23,figure37,figure38 2. Update table1,table2	Yanwu.wang; Xiaobo.bai
2015-10-20	1.03	1. Update table 2 power saving 2. Update table 5 PWRKEY pulled up to 3V 3. Update table 7 of AT+CFUN=1 4. Update table 34 of current consumption 5. Update figure30 and figure31	Yanwu.wang; Xiaobo.bai
2016-07-04	1.04	1. Delete Over-Voltage or Under-Voltage Power off 2. Change PWRKEY pin from at least 1 second to 1.5 second for power off the module 3. Add TVS component to GSM Antenna 4. Add recommended TVS component of GSM Antenna 5. Delete Over-Voltage or Under-Voltage Power off 6. Delete multiplexing function of GPIO 7. Delete Over-Temperature or Under- Temperature Power off 8. Add Recommend SMT stencil footprint	Yanwu.wang; Xiaobo.bai
2017-10-16	1.05	1. Add chapter 8: packaging 2. Modify Chapter 4.5.3	Yue.Hai Ming.zhu

1 Introduction

This document describes SIM800C hardware interface in great detail. The document can help customer to quickly understand SIM800C interface specifications, electrical and mechanical details. With the help of this document and other SIM800C application notes, customer guide, customers can use SIM800C to design various applications quickly.

2 SIM800C Overview

SIM800C is a quad-band GSM/GPRS module that works on frequencies GSM850MHz, EGSM900MHz, DCS1800MHz and PCS1900MHz. SIM800C features GPRS multi-slot class10/class12 (optional) and supports the GPRS coding schemes CS-1, CS-2, CS-3 and CS-4.

With a tiny configuration of 17.6*15.7*2.3mm, SIM800C can meet almost all the space requirements in customers' applications, such as smart phone, PDA and other mobile devices.

SIM800C is a SMT package with 42 pads, and provides all hardware interfaces between the module and customers' boards.

- One 3 lines serial port and one full modem serial port;
- USB interface which can be used for debugging and upgrading firmware;
- One audio channel which include a microphone input and a speaker output;
- Programmable general purpose input and output;
- One SIM card interface;
- Support Bluetooth (need software support).

SIM800C is designed with power saving technique so that the current consumption is as low as 0.6mA in sleep mode.

2.1. SIM800C

Table 1: Module information

	SIM800C
GSM	850,900,1800 and 1900MHz
BT	(need software support)
FLASH	SIM800C (24Mbit) SIM800C32 (32Mbit)
RAM	32Mbit

2.2. SIM800C Key Features

Table 2: SIM800C key features

Feature	Implementation
Power supply	3.4V ~4.4V
Power saving	Typical power consumption in sleep mode is 0.6mA (AT+CFUN=0)
Frequency bands	- Quad-band: GSM 850, EGSM 900, DCS 1800, PCS 1900. SIM800C can search the 4 frequency bands automatically. The frequency bands can also be set by AT command “AT+CBAND”. For details, please refer to <i>document [1]</i>. - Compliant to GSM Phase 2/2+
Transmitting power	- Class 4 (2W) at GSM 850 and EGSM 900 - Class 1 (1W) at DCS 1800 and PCS 1900
GPRS connectivity	- GPRS multi-slot class 12 (default) - GPRS multi-slot class 1~12 (option)
Temperature range	- Normal operation: -40°C ~ +85°C - Storage temperature -45°C ~ +90°C
Data GPRS	- GPRS data downlink transfer: max. 85.6 kbps - GPRS data uplink transfer: max. 85.6 kbps - Coding scheme: CS-1, CS-2, CS-3 and CS-4 - PAP protocol for PPP connect - Integrate the TCP/IP protocol. - Support Packet Broadcast Control Channel (PBCCH)
USSD	Unstructured Supplementary Services Data (USSD) support
SMS	- MT, MO, CB, Text and PDU mode - SMS storage: SIM card
SIM interface	Support SIM card: 1.8V, 3V
External antenna	Antenna pad
Audio features	Speech codec modes: - Half Rate (ETS 06.20) - Full Rate (ETS 06.10) - Enhanced Full Rate (ETS 06.50 / 06.60 / 06.80) - Adaptive multi rate (AMR) - Echo Cancellation - Noise Suppression
Serial port and USB port	Serial port: - Default one Full modem serial port - Can be used for AT commands or data stream - Support RTS/CTS hardware handshake and software ON/OFF flow control - Multiplex ability according to GSM 07.10 Multiplexer Protocol - Autobauding supports baud rate from 1200 bps to 115200bps - upgrading firmware USB port: - USB_DM and USB_DP - Can be used for debugging and upgrading firmware
Phonebook management	Support phonebook types: SM, FD, LD, RC, ON, MC
SIM application toolkit	GSM 11.14 Release 99
Physical characteristics	Size:17.6*15.7*2.3mm Weight:1.3g

Firmware upgrade	Full modern serial port or USB port(recommend to use USB port)
------------------	--

Table 3: Coding schemes and maximum net data rates over air interface

Coding scheme	1 timeslot	2 timeslot	4 timeslot
CS-1	9.05kbps	18.1kbps	36.2kbps
CS-2	13.4kbps	26.8kbps	53.6kbps
CS-3	15.6kbps	31.2kbps	62.4kbps
CS-4	21.4kbps	42.8kbps	85.6kbps

2.3. Operating Mode

The table below summarizes the various operating modes of SIM800C.

Table 4: Overview of operating modes

Mode	Function	
Normal operation	GSM/GPRS SLEEP	Module will automatically go into sleep mode if the conditions of sleep mode are enabling and there aren't on air and hardware interrupt (such as GPIO interrupt or data on serial port). In this case, the current consumption of module will reduce to the minimal level. In sleep mode, the module can still receive paging message and SMS.
	GSM IDLE	Software is active. Module is registered to the GSM network, and the module is ready to communicate.
	GSM TALK	Connection between two subscribers is in progress. In this case, the power consumption depends on network settings such as DTX off/on, FR/EFR/HR, hopping sequences, antenna.
	GPRS STANDBY	Module is ready for GPRS data transfer, but no data is currently sent or received. In this case, power consumption depends on network settings and GPRS configuration.
	GPRS DATA	There is GPRS data transfer (PPP or TCP or UDP) in progress. In this case, power consumption is related with network settings (e.g. power control level); uplink/downlink data rates and GPRS configuration (e.g. used multi-slot settings).
Power off	Normal power off by sending AT command "AT+CPOWD=1" or using the PWRKEY. The power management unit shuts down the power supply for the baseband part of the module. Software is not active. The serial port is not accessible. Power supply (connected to VBAT) remains applied.	
Minimum functionality mode	AT command "AT+CFUN" can be used to set the module to a minimum functionality mode without removing the power supply. In this mode, the RF part of the module will not work or the SIM card will not be accessible, or both RF part and SIM card will be closed, and the serial port is still accessible. The power consumption in this mode is lower than normal	

mode.

2.4. Functional Diagram

The following figure shows a functional diagram of SIM800C:

- GSM baseband
- GSM RF
- Antenna interface
- Other interface

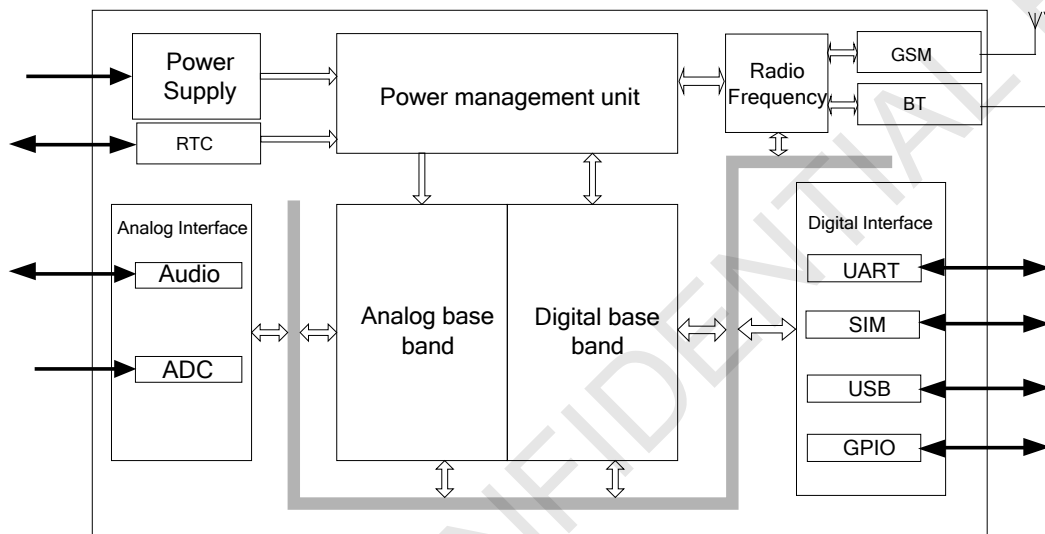


Figure 1: SIM800C functional diagram

3 Package Information

3.1. Pin Out Diagram

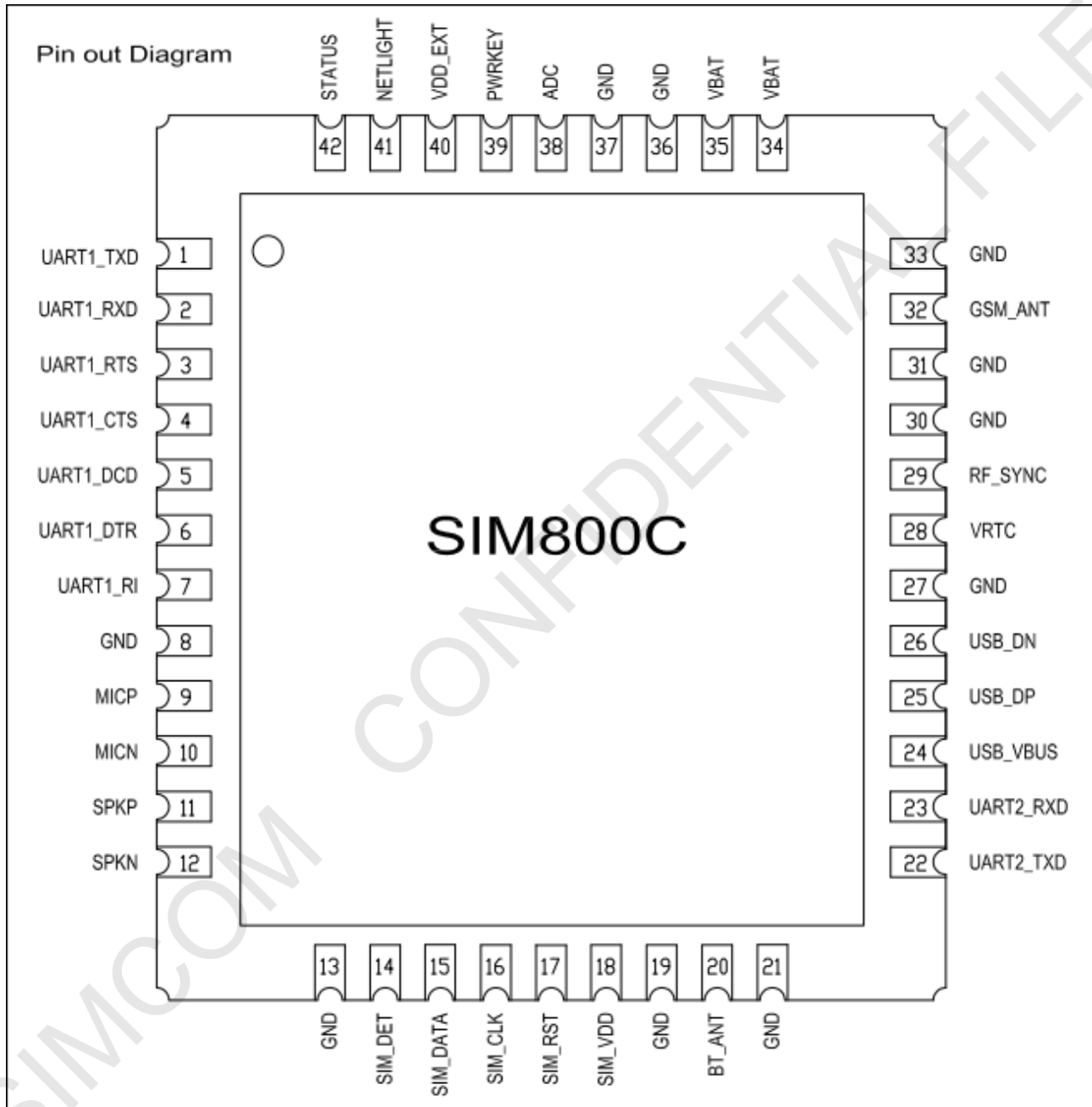


Figure 2: Pin out Diagram (Top view)

3.2. Pin Description

Table 5: Pin description

Pin name	Pin number	I/O	Description	Comment
----------	------------	-----	-------------	---------

Power supply				
VBAT	34,35	I	Power supply	
VRTC	28	I/O	Power supply for RTC	It is recommended to connect with a battery or a capacitor (e.g. 4.7uF).
VDD_EXT	40	O	2.8V power output	If these pins are unused, keep open.
GND	8,13,19,21,27,30,31,33,36,37		Ground	GND for VBAT recommend to use 36,37pin
Power on/down				
PWRKEY	39	I	PWRKEY should be pulled low at least 1 second and then released to power on/down the module.	Internally pulled up to 3V.
Audio interfaces				
MICP	9	I	Differential audio input	If these pins are unused, keep open.
MICN	10			
SPKP	11	O	Differential audio output	
SPKN	12			
GPIO				
NETLIGHT	41	O	Network status	
STATUS	42	O	Power on status	
Serial port				
UART1_DTR	6	I	Data terminal ready	If these pins are unused, keep open.
UART1_RI	7	O	Ring indicator	
UART1_DCD	5	O	Data carrier detect	
UART1_CTS	4	O	Clear to send	
UART1_RTS	3	I	Request to send	
UART1_TXD	1	O	Transmit data	
UART1_RXD	2	I	Receive data	
UART2_TXD	22	O	Transmit data	
UART2_RXD	23	I	Receive data	
Debug interface				
USB_VBUS	24	I	Debug and download	If these pins are unused, keep open.
USB_DP	25	I/O		
USB_DM	26	I/O		
ADC				
ADC	38	I	10bit general analog to digital converter	If these pins are unused, keep open.
SIM card interface				
SIM_VDD	18	O	Voltage supply for SIM card. Support 1.8V or 3V SIM card	All signals of SIM interface should be protected against ESD with a TVS diode array.
SIM_DATA	15	I/O	SIM data input/output	
SIM_CLK	16	O	SIM clock	

SIM_RST	17	O	SIM reset	
SIM_DET	14	I	SIM card detection	If these pins are unused, keep open.
Antenna interface				
GSM_ANT	32	I/O	Connect GSM antenna	
BT_ANT	20	I/O	Connect Bluetooth antenna	
Synchronizing signal of RF				
RF_SYNC	29	O	Synchronizing signal of RF	

3.3. Package Dimensions

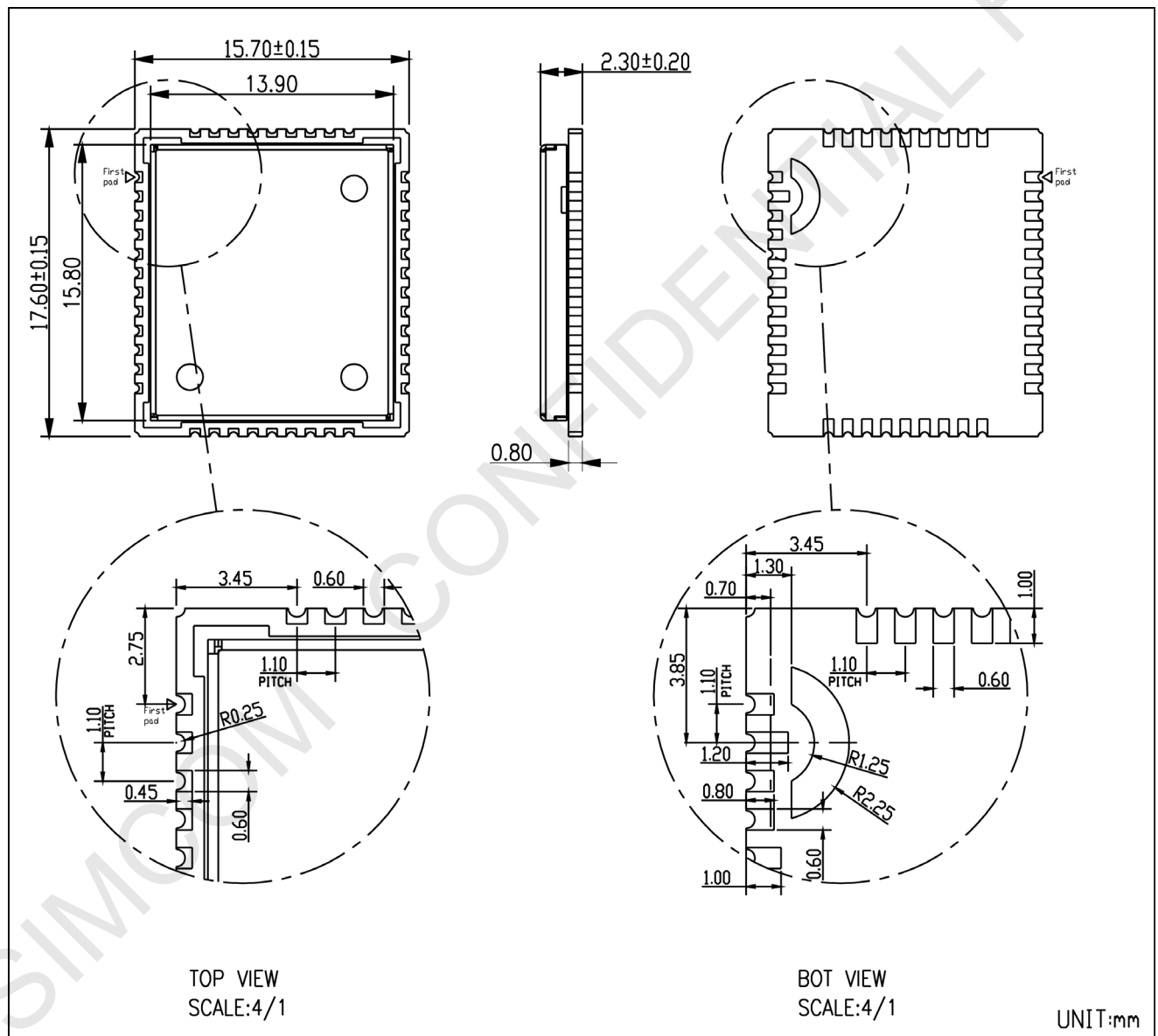


Figure 3: Dimensions of SIM800C (Unit: mm)

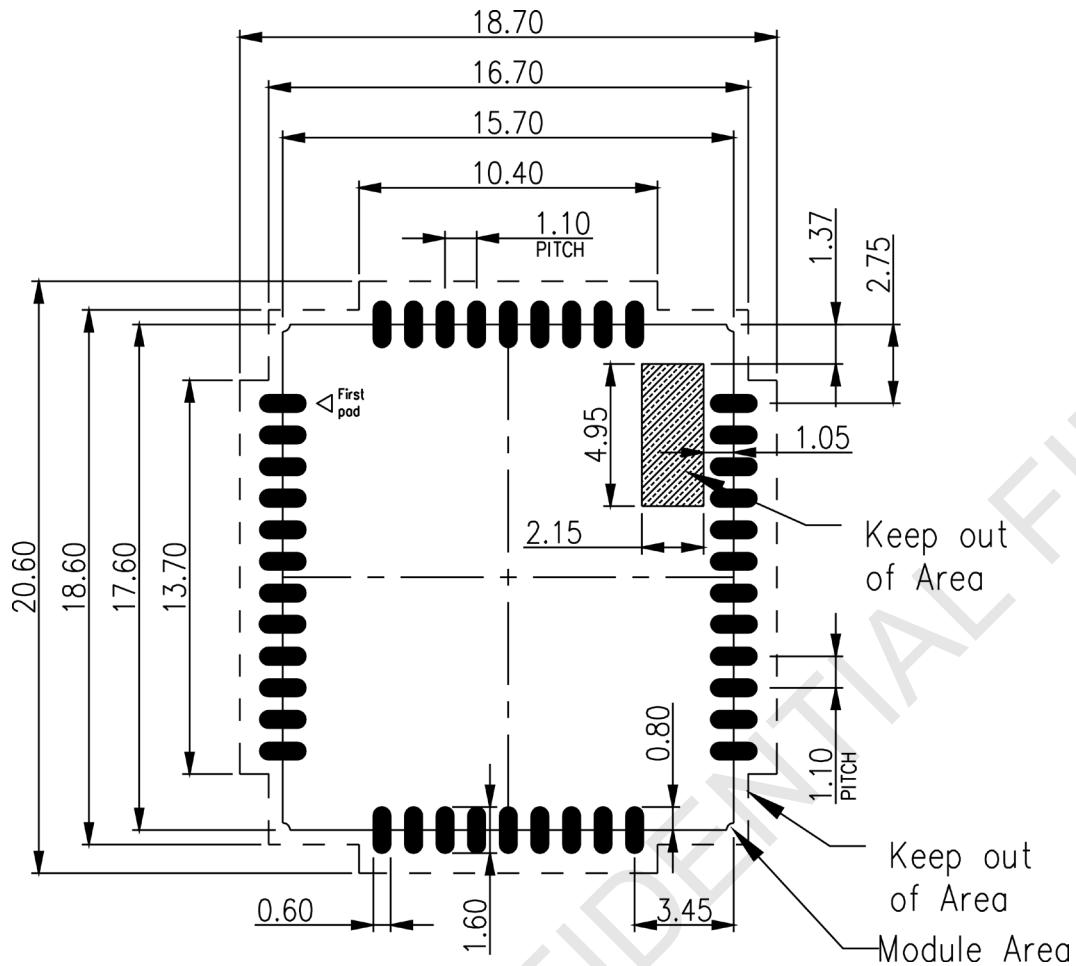


Figure 4: Recommended PCB footprint outline (Unit: mm)

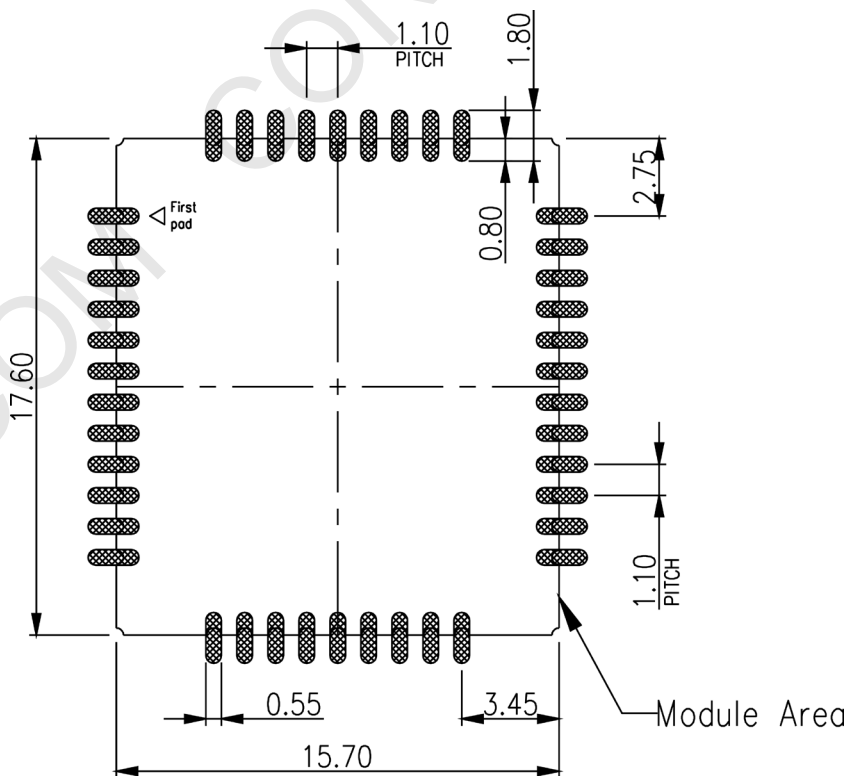


Figure 5: Recommended SMT stencil footprint outline (Unit: mm)

4 Application Interface

4.1. Power Supply

The power supply range of SIM800C is from 3.4V to 4.4V. Recommended voltage is 4.0V. The transmitting burst will cause voltage drop and the power supply must be able to provide sufficient current up to 2A. For the VBAT input, a bypass capacitor (low ESR) such as a 100 μ F is strongly recommended.

For the VBAT input, a 100uF Tantalum capacitor (C_A low ESR) and a 1uF~10uF Ceramics capacitor C_B are strongly recommended. Increase the 33pF and 10pF capacitors can effectively eliminate the high frequency interference. A 5.1V/500mW Zener diode is strongly recommended, the diode can prevent chip from damaging by the voltage surge. These capacitors and Zener diode should be placed as close as possible to SIM800C VBAT pins.

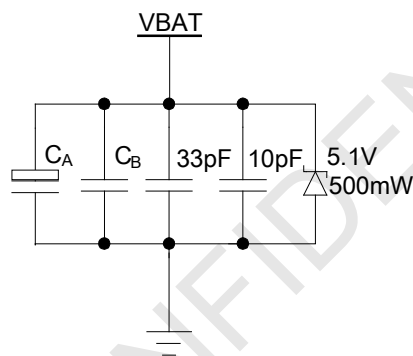


Figure 6: Reference circuit of the VBAT input

Table 6: Recommended zener diode

	Vendor	Part number	Power(watts)	Packages
1	On semi	MMSZ5231BT1G	500mW	SOD123
2	Prisemi	PZ3D4V2H	500mW	SOD323
3	Vishay	MMSZ4689-V	500mW	SOD123
4	Crownpo	CDZ55C5V1SM	500mW	0805

The following figure is the reference design of +5V input power supply. The output power supply is 4.1V, thus a linear regulator can be used.

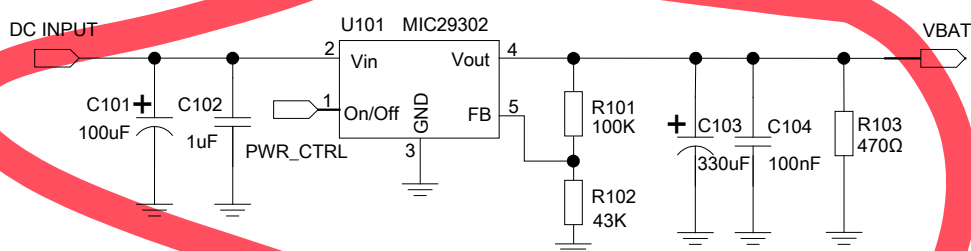


Figure 7: Reference circuit of the LDO power supply

If there is a high drop-out between the input and the desired output (VBAT), a DC-DC power supply will be preferable because of its **better efficiency** especially with the 2A peak current in burst mode of the module. The following figure is the reference circuit.

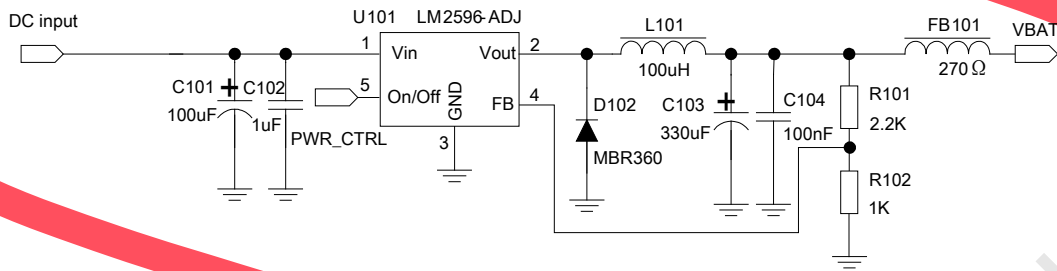


Figure 8: Reference circuit of the DC-DC power supply

The single 3.7V Li-ion cell battery can be connected to SIM800C VBAT pins directly. But the Ni-Cd or Ni-MH battery must be used carefully, since their maximum voltage can rise over the absolute maximum voltage of the module and damage it.

When battery is used, the total impedance between battery and VBAT pins should be less than 150mΩ.

The following figure shows the VBAT voltage drop at the maximum power transmit phase, and the test condition is as following:

VBAT=4.0V,

A VBAT bypass capacitor $C_A=100\mu\text{F}$ tantalum capacitor (ESR=0.7Ω),

Another VBAT bypass capacitor $C_B=1\mu\text{F}\sim 10\mu\text{F}$.

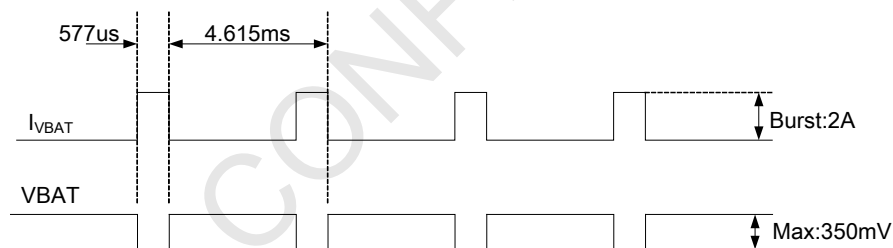


Figure 9: VBAT voltage drop during transmit burst

4.1.1. Power Supply Pin

Pin34 and Pin35 are VBAT input, Pin36 and Pin37 are GND of power supply, and VRTC pin is power supply of the RTC circuit in the module. VDD_EXT output 2.8V when module is in normal operation mode.

When designing the power supply in customers' application, pay special attention to power losses. Ensure that the input voltage never drops below 3.0V even when current consumption rises to 2A in the transmit burst. If the power voltage drops below 3.0V, the module may be shut down automatically. The PCB traces from the VBAT pins to the power supply must be wide enough (at least 60mil) to decrease voltage drops in the transmit burst. The power IC and the bypass capacitor should be placed to the module as close as possible.



Figure 10: The minimal VBAT voltage requirement at VBAT drop

Note: Hardware power off voltage is 3.0V.

4.1.2. Monitoring Power Supply

AT command “AT+CBC” can be used to monitor the VBAT voltage. For detail, please refer to *document [1]*.

4.2. Power on/off SIM800C

4.2.1. Power on SIM800C

Customer can power on SIM800C by pulling down the PWRKEY pin for at least 1 second and release. This pin is already pulled up to 3V in the module internal, so external pull up is not necessary. Reference circuits are shown as below.

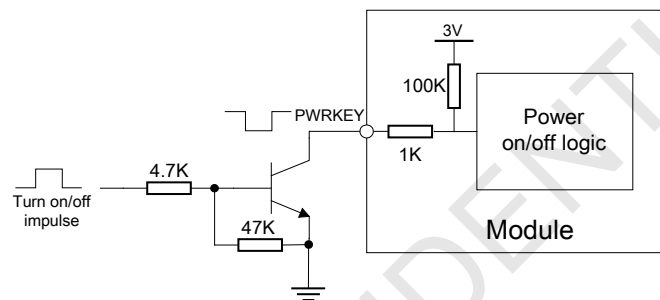


Figure 11: Powered on/down module using transistor

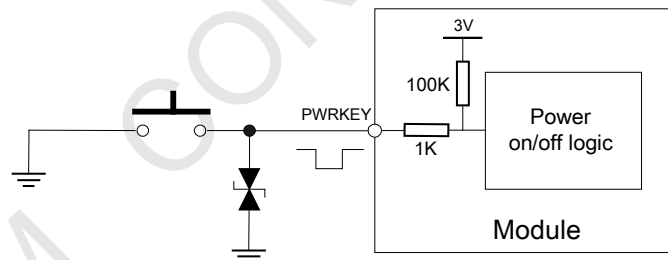


Figure 12: Powered on/down module using button

The power on timing is illustrated as in the following figure.

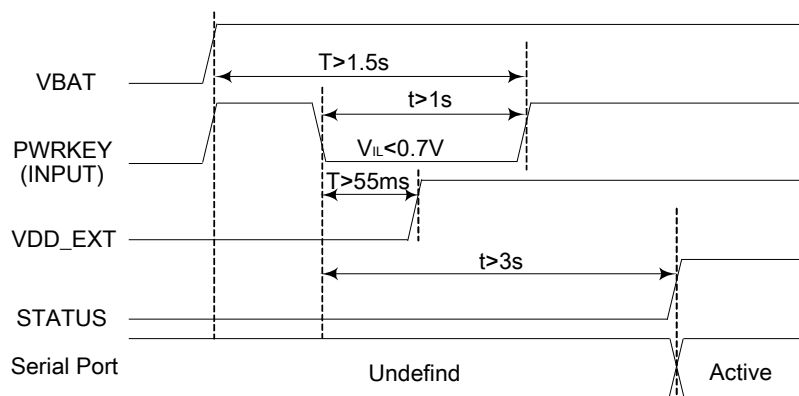


Figure 13: Timing of power on module

When power on procedure is completed, SIM800C will send following URC to indicate that the module is ready to operate at fixed baud rate.

RDY

This URC does not appear when autobauding function is active.

Note: Customer can use AT command “AT+IPR=x” to set a fixed baud rate and save the configuration to non-volatile flash memory. After the configuration is saved as fixed baud rate, the Code “RDY” should be received from the serial port every time when SIM800C is powered on. For details, please refer to the chapter “AT+IPR” in document [1].

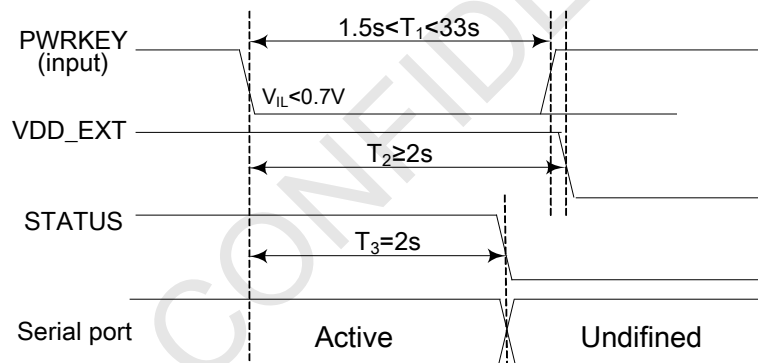
4.2.2. Power off SIM800C

SIM800C will be powered off in the following situations:

- Normal power off procedure: power off SIM800C by the PWRKEY pin.
- Normal power off procedure: power off SIM800C by AT command “AT+CPOWD=1”.

4.2.2.1. Power off SIM800C by the PWRKEY Pin

Customer can power off SIM800C by pulling down the PWRKEY pin for at least 1.5 second and release. Please refer to the power on circuit. The power off timing is illustrated in the following figure.


Figure 14: Timing of power off SIM800C by PWRKEY

Note:

1. the module will restart after pull down the pwrkey over 33 seconds.
2. VDD_EXT will power off after STATUS change into low level and the PWRKEY release 55ms..

If $1.5s < T_1 < 2s$, $T_2 > 2s$;

If $2s \leq T_1 < 33s$, $T_2 > T_1 + 55ms$

This procedure makes the module log off from the network and allows the software to enter into a secure state to save data before completely shut down.

Before the completion of the power off procedure, the module will send URC:

NORMAL POWER OFF

At this moment, AT commands can not be executed any more. Power off mode can also be indicated by STATUS pin, which is at low level at this time.

4.2.2.2. Power off SIM800C by AT Command

SIM800C can be powered off by AT command “AT+CPOWD=1”. This procedure makes the module log off from the network and allows the software to enter into a secure state to save data before completely shut down.

Before the completion of the power off procedure, the module will send URC:

NORMAL POWER OFF

At this moment, AT commands can not be executed any more. Power off mode can also be indicated by STATUS pin, which is at low level at this time.

For detail about AT command “AT+CPOWD”, please refer to *document [1]*.

4.2.2.3. Restart SIM800C by PWRKEY Pin:

When the module works normally, if the customer wants to restart the module, follow the procedure below:

- 1) Power off the module.
- 2) Wait for at least 800ms after STATUS pin changed to low level.
- 3) Power on the module.

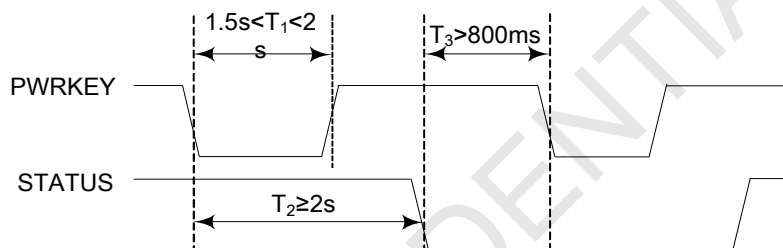


Figure 15: Timing of restart SIM800C

4.3. Power Saving Mode

SIM800C has two power saving modes: Minimum functionality mode and sleep mode. AT command “AT+CSCLK=1” can be used to set SIM800C into sleep mode. AT command “AT+CFUN=<fun>” can be used to set SIM800C into minimum functionality. When SIM800C is in sleep mode and minimum functionality mode, the current of module is lowest.

4.3.1. Minimum Functionality Mode

There are three functionality modes, which could be set by AT command “AT+CFUN=<fun>”. The command provides the choice of the functionality levels <fun>=0, 1, 4.

- AT+CFUN=0: Minimum functionality.
- AT+CFUN=1: Full functionality (default).
- AT+CFUN=4: Flight mode (disable RF function).

Table 7: The current consumption of Function Mode

<fun>	BS-PA-MFRMS	Current consumption(mA)
0	/	0.6
1	9	0.9
	5	1
	2	1.5

Minimum functionality mode minimizes the current consumption to the lowest level. If SIM800C is set to minimum functionality by “AT+CFUN=0”, the RF function and SIM card function will be disabled. In this case, the serial port is still accessible, but partial AT commands and correlative to RF function and SIM card function will not be accessible.

For detailed information about AT command “AT+CFUN=<fun>“, please refer to *document [1]*.

4.3.2. Sleep Mode 1 (AT+CSCLK=1)

Customer can control SIM800C module to enter or exit the sleep mode (AT+CSCLK=1) by DTR signal. When DTR is in high level and without interrupt (on air and hardware such as GPIO interrupt or data in serial port), SIM800C will enter sleep mode automatically. In this mode, SIM800C can still receive paging or SMS from network but the serial port is not accessible.

4.3.3. Wake Up SIM800C from Sleep Mode 1

When SIM800C is in sleep mode 1(AT+CSCLK=1), the following methods can wake up the module:

- Pull down DTR pin.
The serial port will be active after DTR pin is pulled to low level for about 50ms.
- Receive a voice or data call from network.
- Receive a SMS from network.
- Receive external interrupt.

Note: After module has received incoming call or new SMS, serial port can report URC, but the serial port can not input AT command. Only after the DTR pin is pulled to low level for 50ms, the serial port can input AT command.

4.3.4. Sleep Mode 2 (AT+CSCLK=2)

In this mode, SIM800C will continuously monitor the serial port data signal. When there is no data transfer over 5 seconds on the RXD signal and there is no on air and hardware interrupts (such as GPIO interrupt), SIM800C will enter sleep mode 2 automatically. In this mode, SIM800C can still receive paging or SMS from network.

4.3.5. Wake Up SIM800C from Sleep Mode 2

When SIM800C is in sleep mode 2 (AT+CSCLK=2), the following methods can wake up the module:

- Send data to SIM800C via main serial port (the first character will lose).
- Receive a voice or data call from network.
- Receive a SMS from network.

Note: Autobauding is default. It cannot enter sleep mode in the absence of synchronous serial port baud rate after module power on.

4.4. Power Saving Mode

Current input for RTC when the VBAT is not supplied for the system. Current output for backup battery when the VBAT power supply is in present and the backup battery is in low voltage state. The RTC power supply of

module can be provided by an external capacitor or a battery (non-chargeable or rechargeable) through the VRTC. The following figures show various reference circuits for RTC back up.

- External capacitor backup

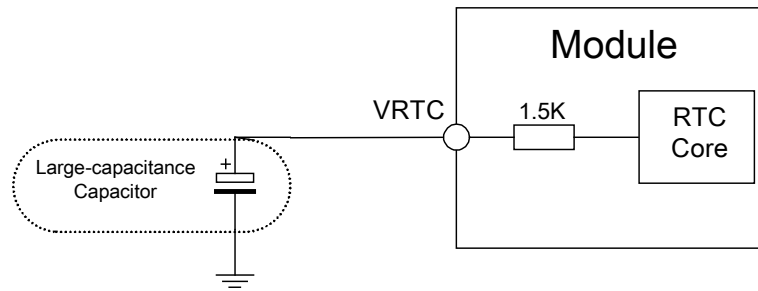


Figure 16: RTC supply from capacitor

- Non-chargeable battery backup

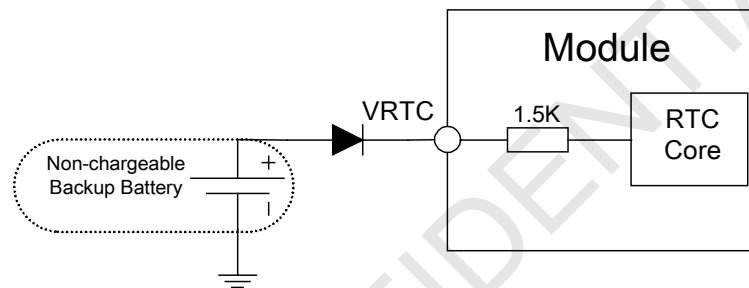


Figure 17: RTC supply from non-chargeable battery

- Rechargeable battery backup

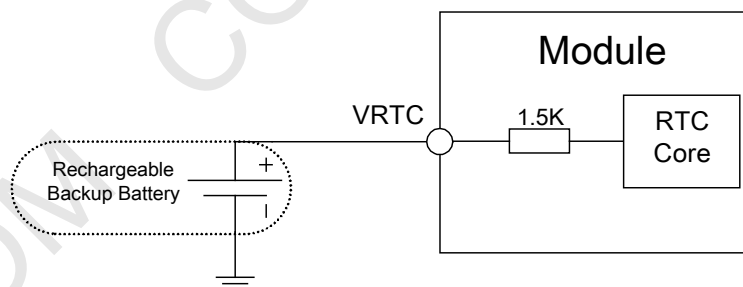


Figure 18: RTC supply from rechargeable battery

Note: When shut off VBAT and power on VRTC only, the clock error becomes larger.

4.5. Serial Port and USB Interface

SIM800C default provides one unbalanced asynchronous serial ports. The module is designed as a DCE (Data Communication Equipment). The following figure shows the connection between module and client (DTE).

Table 8: Serial port and USB pin definition

	Pin name	Pin number	Function
--	----------	------------	----------

Serial port	UART1_DTR	6	Data terminal ready
	UART1_RI	7	Ring indicator
	UART1_DCD	5	Data carrier detect
	UART1_CTS	4	Clear to send
	UART1_RTS	3	Request to send
	UART1_TXD	1	Transmit data
	UART1_RXD	2	Receive data
	UART2_TXD	22	Transmit data
Debug port	UART2_RXD	23	Receive data
	USB_VBUS	24	USB power supply
	USB_DP	25	D+ data input/output
	USB_DM	26	D- data input/output

Note: Hardware flow control is disabled by default. AT command “AT+IFC=2, 2” can enable hardware flow control. AT command “AT+IFC=0,0” can disable hardware flow control. For more details please refer to document [1].

Table 9: Serial port characteristics

Symbol	Min	Max	Unit
V _{IL}	-0.3	0.7	V
V _{IH}	2.1	3.0	V
V _{OL}	-	0.4	V
V _{OH}	2.4	-	V

4.5.1 Function of Serial Port

Serial port:

- Full mode device.
- Contain data lines UART1_TXD/UART1_RXD, hardware flow control lines UART1_RTS/UART1_CTS, status lines UART1_DTR、UART1_DCD and UART1_RI.
- Serial port can be used for GPRS service and AT communication. It can also be used for multiplexing function. For details about multiplexing function, please refer to table 11.
- Autobauding supports the following baud rates:
1200, 2400, 4800, 9600, 19200, 38400, 57600 and 115200bps

Autobauding allows SIM800C to automatically detect the baud rate of the host device. Pay more attention to the following requirements:

- **Synchronization between DTE and DCE:**
When DCE powers on with autobauding enabled, it is recommended to send "AT" or "at" or "aT" or "At" to synchronize the baud rate, until DTE receives the "OK" response, which means DTE and DCE are correctly synchronized. For more information please refer to AT command "AT+IPR".
- **Restrictions of autobauding operation:**
The DTE serial port must be set at 8 data bits, no parity and 1 stop bit.
The URC such as "RDY", "+CFUN: 1" and "+CPIN: READY" will not be reported.

Note: Customer can use AT command “AT+IPR=x” to set a fixed baud rate and the setting will be saved to non-volatile flash memory automatically. After the configuration is set as fixed baud rate, the URC such as “RDY”, “+CFUN: 1” and “+CPIN: READY” will be reported when SIM800C is powered on.

4.5.2 Serial Interfaces

The following figure shows the connection between module and client (DTE).

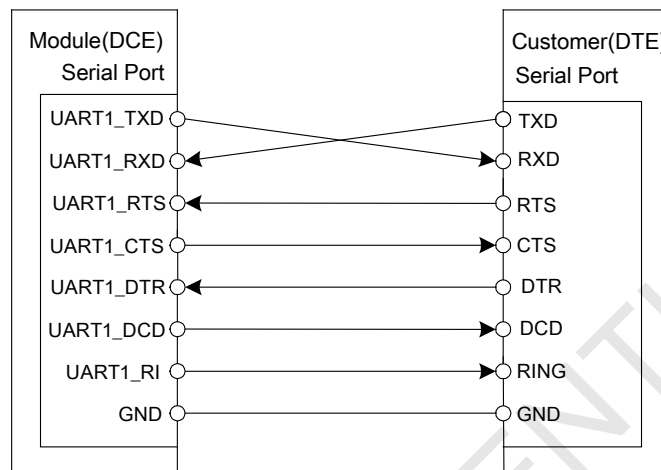


Figure 19: Connection of the serial interfaces

If the voltage of UART is 3.3V, the following reference circuits are recommended. If the voltage is 3.0V, please change the resistors in the following figure from 5.6K to 14K.

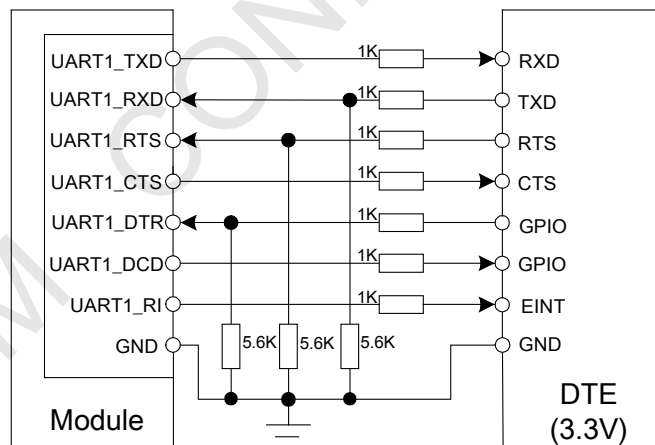


Figure 20: Resistor matching circuit

If the voltage of UART is 3V or 3.3V, the following reference circuits are recommended:

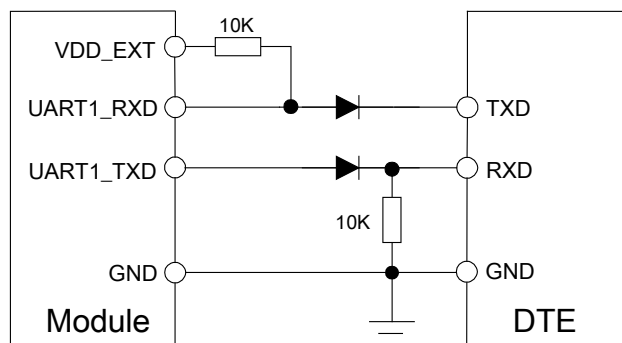
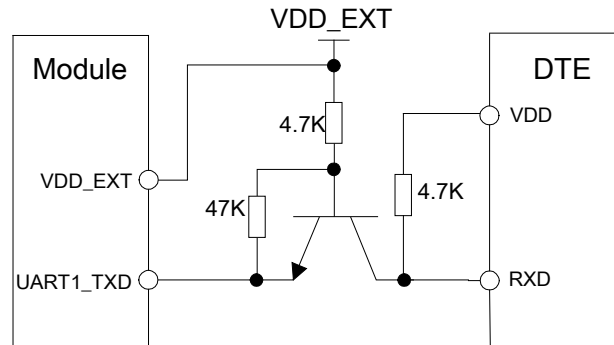
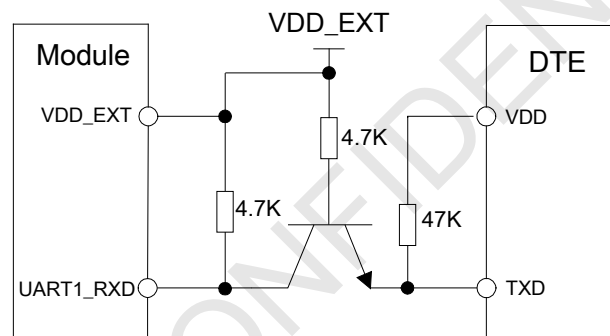


Figure 21 : Diode isolation circuit

Note: please make sure the minimum of client high limit should be less than 2.8V minus the diode drop.

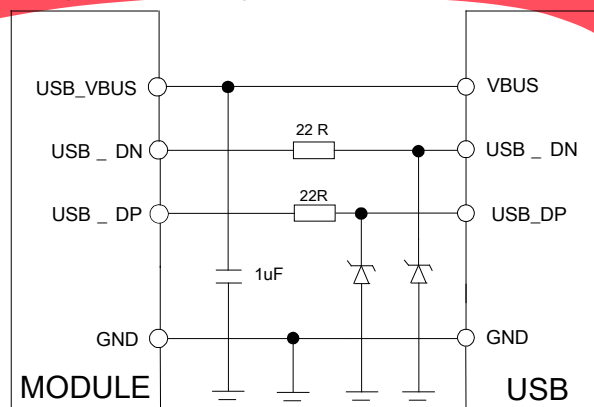
If the voltage of UART is 5V, the following reference circuits are recommended:


Figure 22: TX level matching circuit

Figure 23: RX level matching circuit

4.5.3 Debug Interface

SIM800C could achieve software debug function through USB interface. When powering on the module, connect USB_VBUS, USB_DP, USB_DN, and GND to PC, then install the driver following the prompts, a UART port could be recognized by PC, customer could achieve the software Debug with this UART port.

SIMCom recommended the following connected diagram:


Figure 24: USB reference circuit

The TVS on USB data line should be less than 5pF, and traced by differential forms.

Note: please reserve the USB interface or test point for the further debugging

Table 10: USB_VBUS operation voltage

Pin	Min	Typ	Max	Unit
USB_VBUS	4.3	5.0	7.0	V

4.5.4 Software Upgrade

Customer could upgrade module's firmware through USB or UART interface.

If upgrading through USB interface, it is necessary to power on SIM800C first, and then connect USB_VBUS, USB_DP, USB_DM, and GND to PC. There is no need to operate PWRKEY pin in the whole procedure, when SIM800C detects USB_VBUS and could communicate normally with USB_DP and USB_DM, it will enter USB download mode automatically.

If customer upgrades the software through UART interface, it is strongly recommended to lead the UART1_TXD, UART1_RXD, GND and PWRKEY pin to IO connector for the upgrading, and PWRKEY pin should connect to GND while upgrading. Refer to the following figure for debugging and upgrading software.

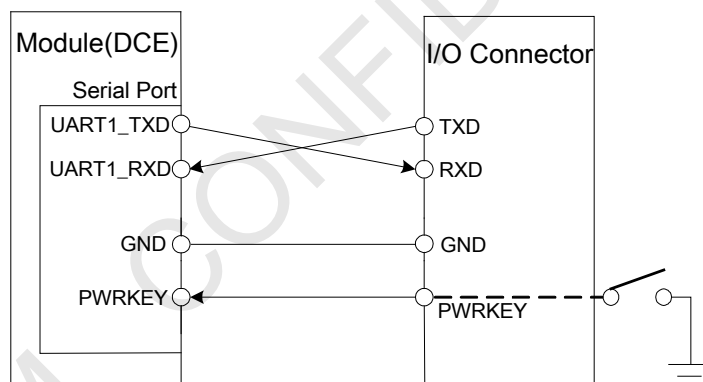


Figure 25: Connection for software upgrading and debugging

The UART interface supports the CMOS level. If customer connects the module to the computer, the level shifter should be added between the DCE and DTE.

4.6. UART1_RI Behaviors

Table 11: RI behaviors

State	RI response
Standby	High
Voice call	The pin is changed to low. When any of the following events occur, the pin will be changed to high: (1) Establish the call (2) Hang up the call
SMS	The pin is changed to low, and kept low for 120ms when a SMS is received. Then it is changed

	to high.
Others	For more details, please refer to <i>document [2]</i> .

The behavior of the RI pin is shown in the following figure when the module is used as a receiver.

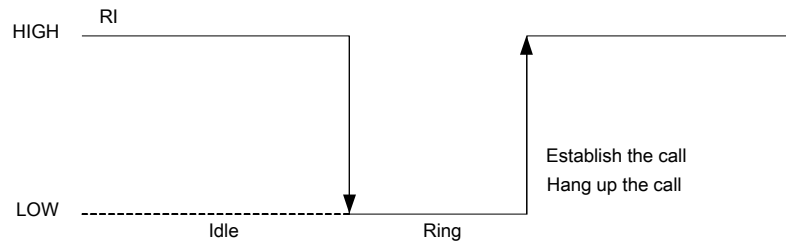


Figure 26: UART1_RI behaviour of voice calling as a receiver

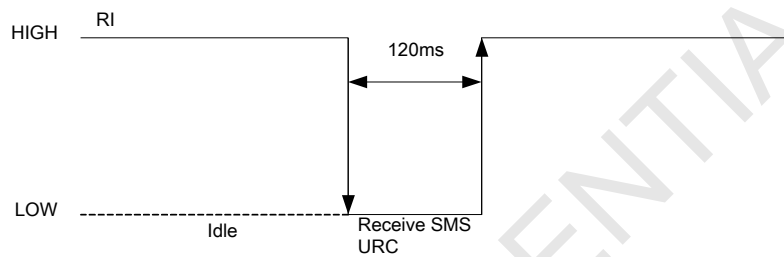


Figure 27: UART1_RI behaviour of URC or receive SMS

However, if the module is used as caller, the UART1_RI will remain high. Please refer to the following figure.



Figure 28: UART1_RI behaviour as a caller

4.7. Audio Interfaces

SIM800C provides an analog input (MICP; MICN), which could be used for electret microphone. The module also provides an analog output (SPKP; SPKN).

Table 12: Audio interface definition

Pin name	Pin number	Function
MICP	9	Audio input positive
MICN	10	Audio input negative
SPKP	11	Audio output positive
SPKN	12	Audio output negative

SPKP/SPKN output can directly drive 32Ω receiver.

AT command “AT+CMIC” is used to adjust the input gain level of microphone. AT command “AT+SIDET” is used to set the side-tone level. In addition, AT command “AT+CLVL” is used to adjust the output gain level. For more details, please refer to *document [1]*.

In order to improve audio performance, the following reference circuits are recommended. The audio signals have to be layout according to differential signal layout rules as shown in following figures.

4.7.1. Speaker Interfaces Configuration

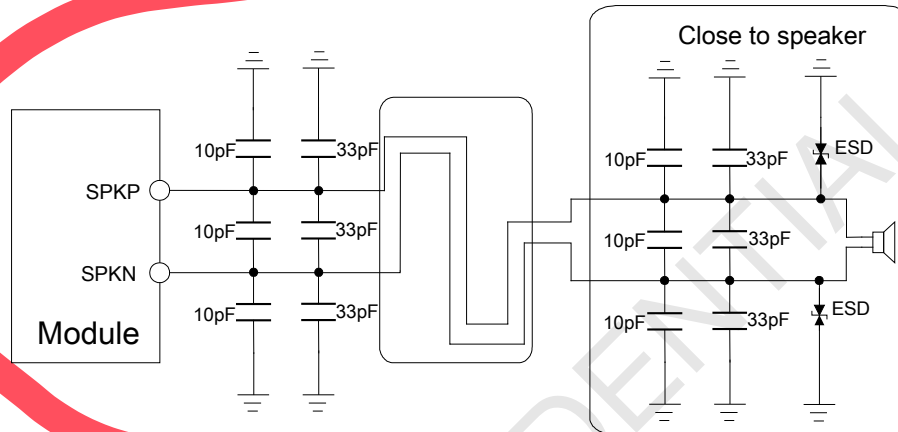


Figure 29: Speaker reference circuit

4.7.2. Microphone Interfaces Configuration

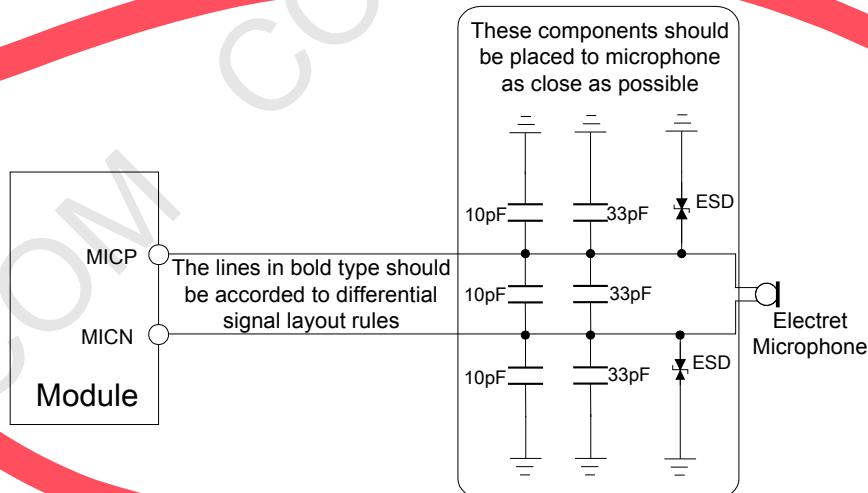


Figure 30: Microphone reference circuit

4.7.3. Audio Electronic Characteristic

Table 13: Microphone input characteristics

Parameter	Min	Typ	Max	Unit
Microphone biasing voltage	-	1.9	2.2	V

Working current	-	-	2.0	mA
Input impedance(differential)	13	20	27	KΩ
Idle channel noise	-	-	-67	dBm0
SINAD	Input level:-40dBm0	29	-	dB
	Input level:0dBm0	-	69	dB

Table 14: Audio output characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Normal output	R _L =32 Ω receiver	-	15	90	mW

4.7.4. TDD

Audio signal could be interferenced by RF signal. Coupling noise could be filtered by adding 33pF and 10pF capacitor to audio lines. 33pF capacitor could eliminate noise from GSM850/EGSM900MHz, while 10pF capacitor could eliminate noise from DCS1800/PCS1900Mhz frequency. Customer should develop this filter solution according to field test result.

GSM antenna is the key coupling interfering source of TDD noise. Thereat, pay attention to the layout of audio lines which should be far away from RF cable, antenna and VBAT pin. The bypass capacitor for filtering should be placed near module and another group needs to be placed near to connector.

Conducting noise is mainly caused by the VBAT drop. If audio PA was powered by VBAT directly, then there will be some cheep noise from speaker output easily. So it is better to put big capacitors and ferrite beads near audio PA input.

TDD noise has something to do with GND signal. If GND plane is not good, lots of high-frequency noises will interference microphone and speaker over bypass capacitor. So a good GND during PCB layout could avoid TDD noise.

4.8. SIM Card Interface

The SIM interface complies with the GSM Phase 1 specification and the new GSM Phase 2+ specification for FAST 64kbps SIM card. Both 1.8V and 3.0V SIM card are supported. The SIM interface is powered from an internal regulator in the module.

4.8.1. SIM Card Application

Table 15: SIM pin definition

Pin name	Pin number	Function
SIM_VDD	18	Voltage supply for SIM card. Support 1.8V or 3V SIM card
SIM_DATA	15	SIM data input/output
SIM_CLK	16	SIM clock
SIM_RST	17	SIM reset
SIM_DET	14	SIM card detection

It is recommended to use an ESD protection component such as ST (www.st.com) ESDA6V1-5W6. The SIM card peripheral components should be placed close to the SIM card holder. The reference circuit of the 8-pin SIM card holder is illustrated in the following figure.

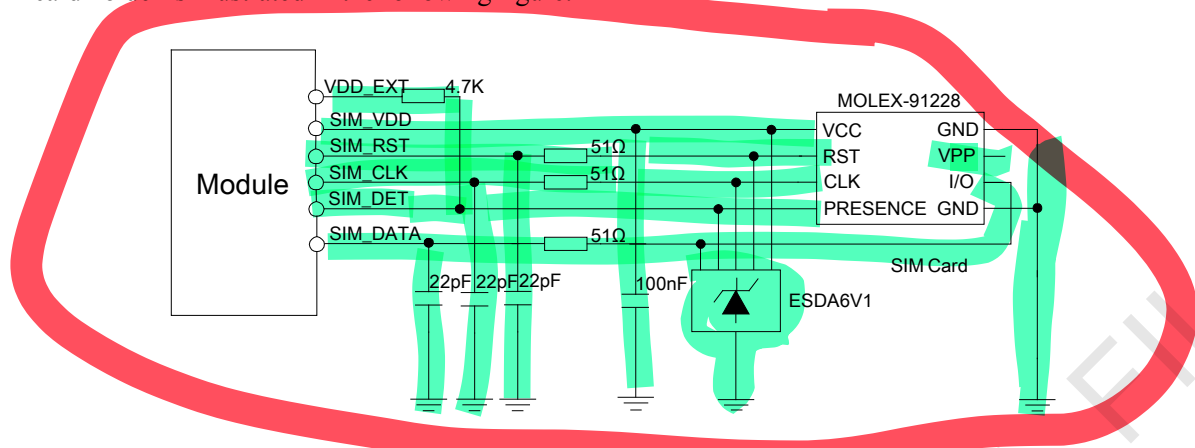


Figure 31: Reference circuit of the 8-pin SIM card holder

The SIM_DET pin is used for detection of the SIM card hot plug in. Customer can select the 8-pin SIM card holder to implement SIM card detection function. AT command “AT+CSDT” is used to enable or disable SIM card detection function. For details of this AT command, please refer to *document [1]*.

If the SIM card detection function is not used, customer can keep the SIM_DET pin open. The reference circuit of 6-pin SIM card holder is illustrated in the following figure.

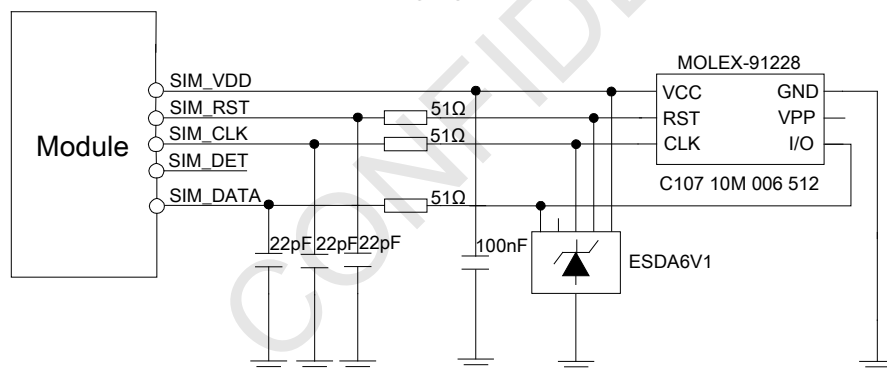


Figure 32: Reference circuit of the 6-pin SIM card holder

4.8.2. SIM Card Design Guide

SIM card signal could be interferenced by some high frequency signal, it is strongly recommended to follow these guidelines while designing:

- SIM card holder should be far away from GSM antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIM card holder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 51Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESD protection
- Add 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal to induce RF signal interference

4.8.3. Design Considerations for SIM Card Holder

For 8 pins SIM card holder, SIMCom recommends to use Molex 91228. Customer can visit <http://www.molex.com> for more information about the holder.

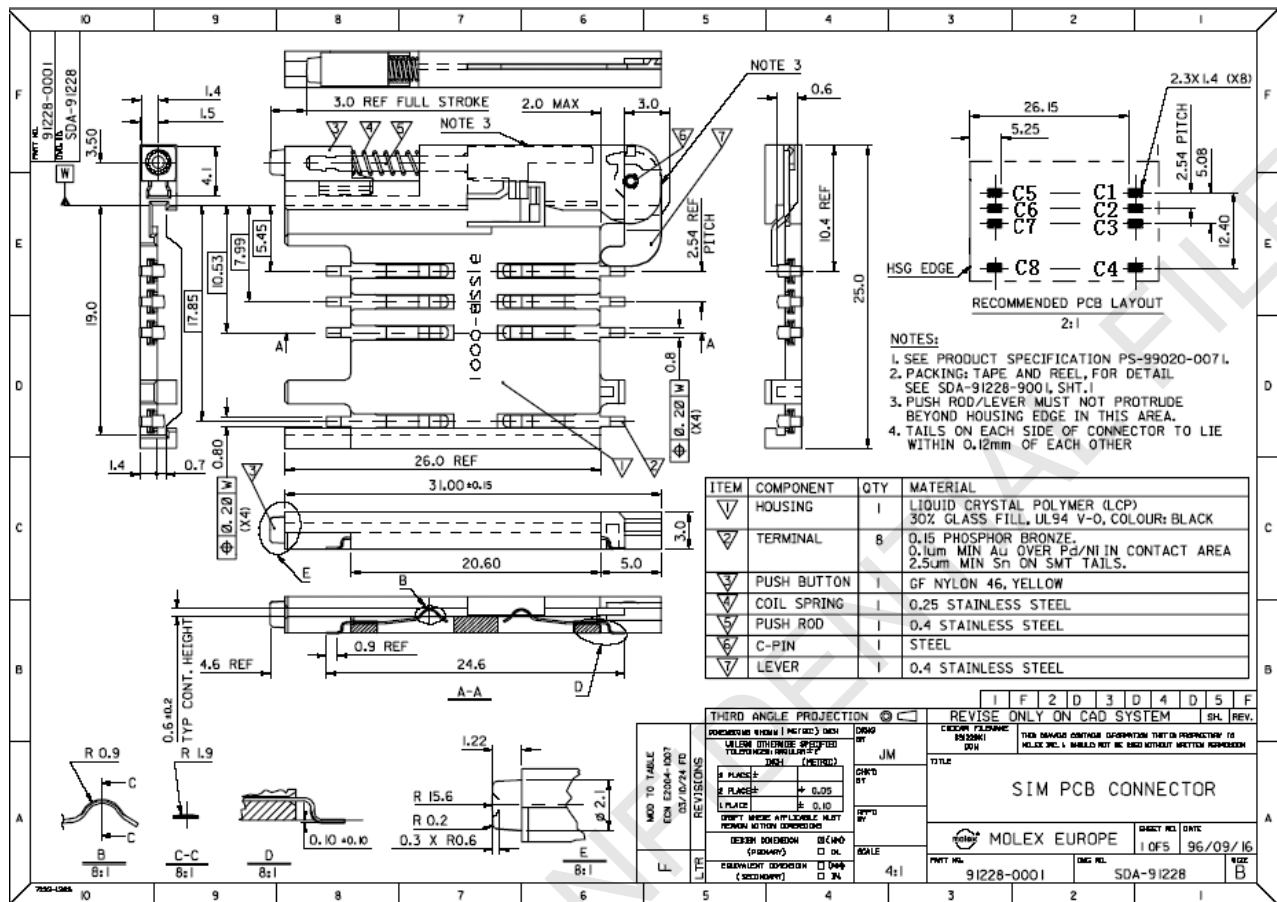


Figure 33: Molex 91228 SIM card holder

Table 16: Pin description (Molex SIM card holder)

Pin name	Signal	Description
C1	SIM_VDD	SIM card power supply
C2	SIM_RST	SIM card reset
C3	SIM_CLK	SIM card clock
C4	GND	Connect to GND
C5	GND	Connect to GND
C6	VPP	Not connect
C7	SIM_DATA	SIM card data I/O
C8	SIM_DET	Detect SIM card presence

For 6-pin SIM card holder, SIMCom recommends to use Amphenol C707 10M006 512 .Customer can visit <http://www.amphenol.com> for more information about the holder.

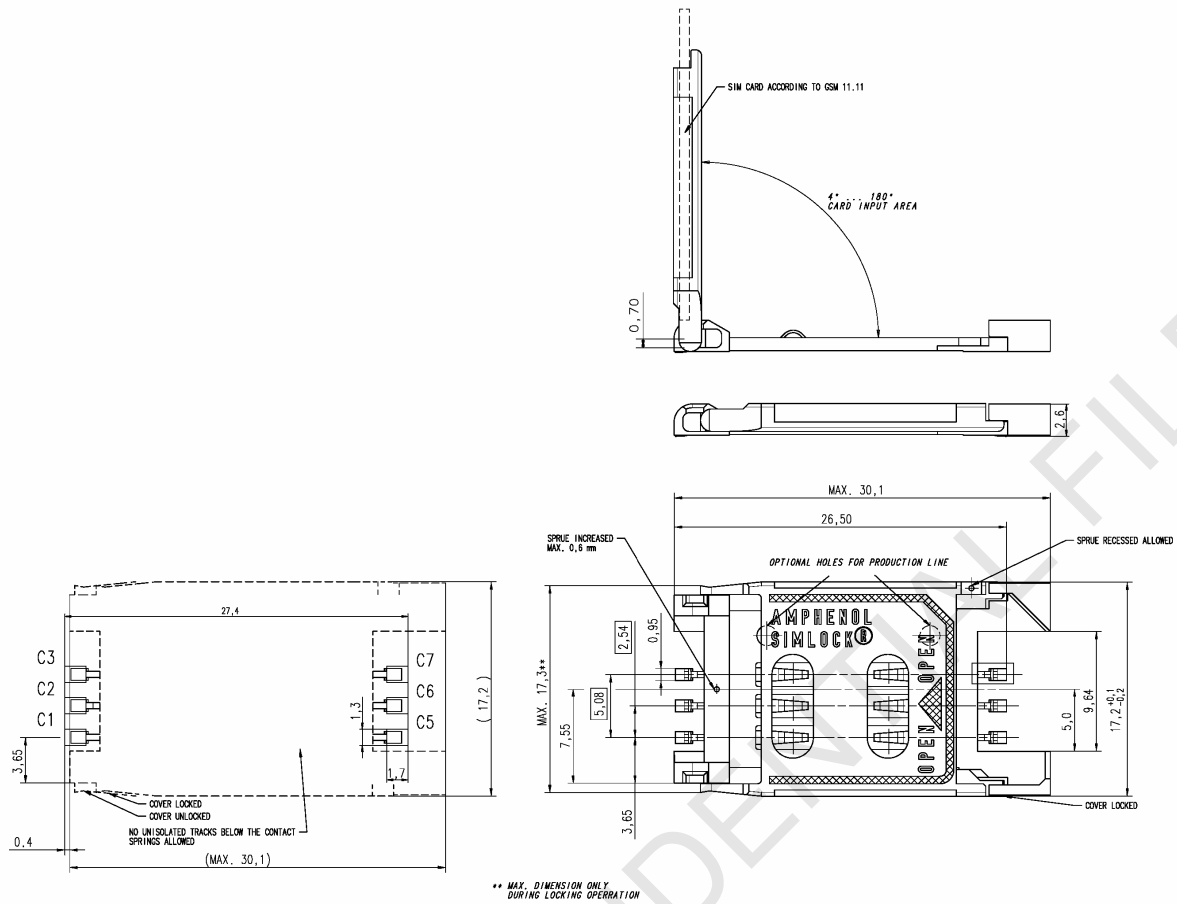


Figure 34: Amphenol C707 10M006 512 SIM card holder

Table 17: Pin description (Amphenol SIM card holder)

Pin name	Signal	Description
C1	SIM_VDD	SIM card power supply
C2	SIM_RST	SIM card reset
C3	SIM_CLK	SIM card clock
C5	GND	Connect to GND
C6	VPP	Not connect
C7	SIM_DATA	SIM card data I/O

Note: Every time plug SIM card interval advice is greater than 2s. Otherwise may not be able to correct detection.

4.9. ADC

Table 18: Pin definition of the ADC

Pin name	Pin number	Description
ADC	38	Analog voltage input

SIM800C provides an auxiliary ADC, which can be used to measure the voltage. Customer can use AT command “AT+CADC” to read the voltage value. For details of this AT command, please refer to **document [1]**.

Table 19: ADC specification

Parameter	Min	Typ	Max	Unit
Voltage range	0	-	2.8	V
ADC Resolution	-	10	-	bits
Sampling rate	-	-	1.0833	MHz
ADC precision		10	30	mV

4.10. Network Status Indication

Table 20: Pin definition of the NETLIGHT

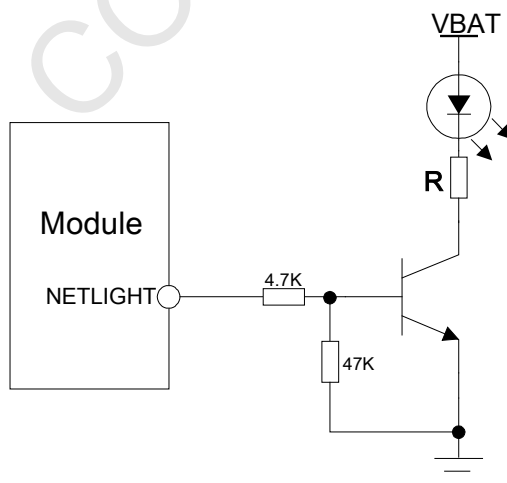
Pin name	Pin number	Description
NETLIGHT	41	Network Status Indication

The NETLIGHT pin can be used to drive a network status indication LED. The status of this pin is listed in following table:

Table 21: Status of the NETLIGHT pin

Status	SIM800C behavior
Off	Powered off
64ms On/ 800ms Off	Not registered the network
64ms On/ 3000ms Off	Registered to the network
64ms On/ 300ms Off	GPRS communication is established

Reference circuit is recommended in the following figure:


Figure 35: Reference circuit of NETLIGHT

4.11. Operating Status Indication

The pin42 is for operating status indication of the module. The pin output is high when module is powered on, and output is low when module is powered off.

Table 22: Pin definition of the STATUS

Pin name	Pin number	Description
STATUS	42	Operating status indication

Note: For timing about STATUS, please reference to the chapter “4.2 power on/down scenarios”

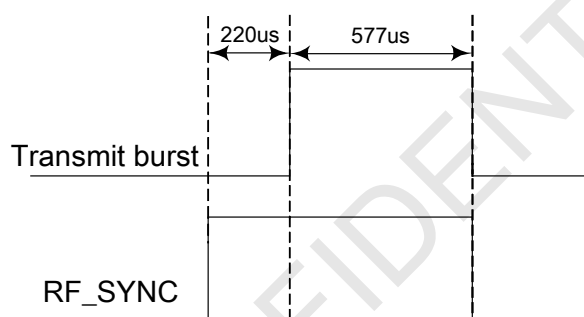
4.12. RF Synchronization Signal

The synchronization signal is used to indicate incoming GSM burst.

Table 23: Definition of the RF_SYNC pin

Pin name	Pin number	Description
RF_SYNC	29	Transmit synchronization signal

The timing of the synchronization signal is shown below.


Figure 36: RF_SYNC signal during transmit burst

4.13. Antenna Interface

There are two antenna interfaces, GSM_ANT and BT_ANT.

- The input impedance of the antenna should be 50Ω, and the VSWR should be less than 2.
- It is recommended that the GSM antenna and the BT antenna should be placed as far as possible.
- The isolations of the two antenna should be bigger than 30dB

NOTE: About the RF trace layout please refer to “AN_SMT Module_RF_Reference Design_Guide”.

4.13.1 GSM Antenna Interface

SIM800C provides a GSM antenna named GSM_ANT, customer could use 50Ω microstrip line or stripline antenna connect to the module.

It is recommended to reserve the matching circuit as following:

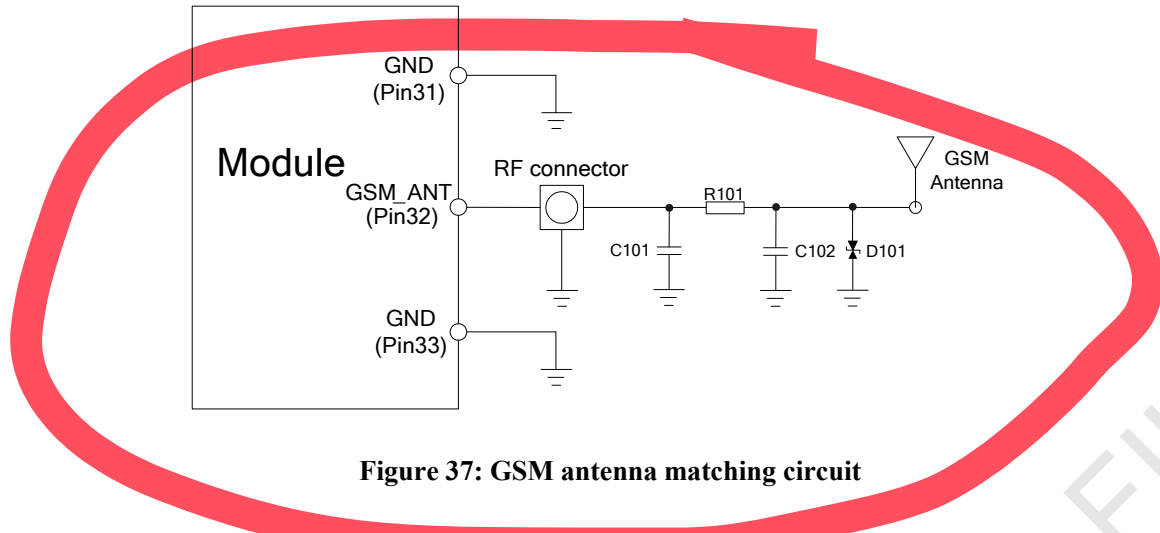


Figure 37: GSM antenna matching circuit

R101, C101, C102 are the matching circuit, the value should be defined by the antenna design. Normally R101 is 0Ω , C101 and C102 are not mounted.

The RF connector is used for conduction test. If the space between RF pin and antenna is not enough, the matching circuit should be designed as in the following figure:

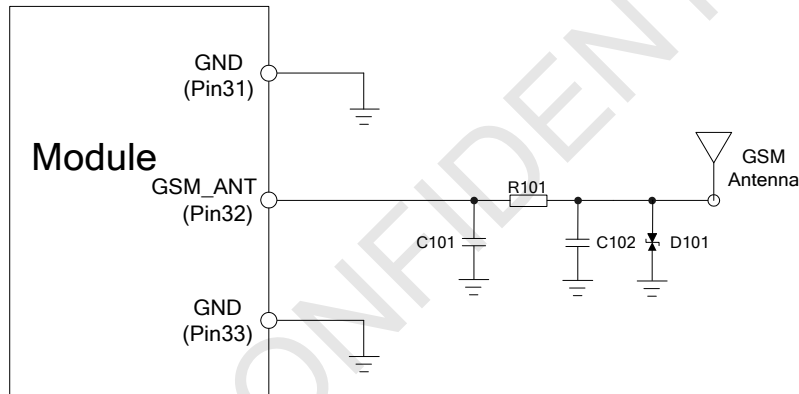


Figure 38: GSM antenna matching circuit without RF connector

In above figure, the components R101, C101 and C102 are used for antenna matching, the value of components can only be got after the antenna tuning, usually, they are provided by antenna vendor. By default, the R101 is 0Ω resistors, and the C101, C102 are reserved for tuning.

The RF test connector in the figure is used for the conducted RF performance test, and should be placed as close as to the module's antenna pin. The traces impedance between components must be controlled in 50Ω . The component D101 is a bidirectional TVS component, which is used for ESD protection, the recommended part numbers of the TVS are listed in the following table:

Table 24: Recommended TVS component

Package	Type	Supplier
0201	LXES03AAA1-098	Murata
0201	LXES03AAA1-154	Murata
0402	LXES15AAA1-153	Murata
0402	LXES15AAA1-100	Murata
0402	LXES15AAA1-017	Murata

4.13.2 Bluetooth Antenna Interface

It is recommended to reserve the matching circuit as following:

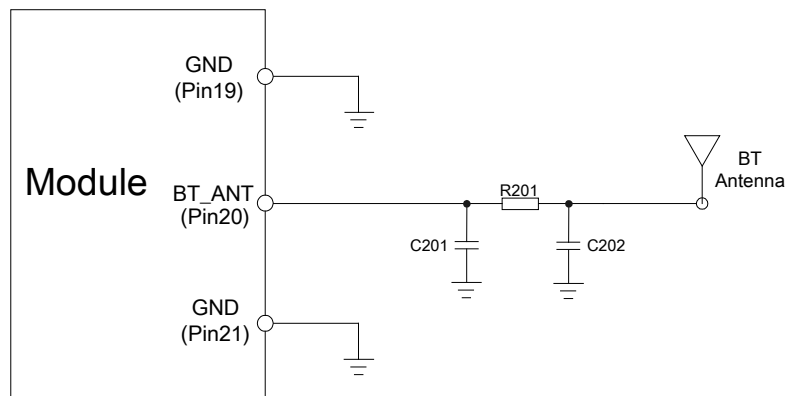


Figure 39: Bluetooth antenna matching circuit

R201, C201, C202 are the matching circuit, the value should be defined by the antenna design. Normally R201 is 0Ω, C202 and C201 are not mounted.

There are some suggestions for placing components and RF trace for GSM_ANT/BT_ANT:

- The RF connector is used for conducted test, so keep it as close to pin GSM_ANT as possible;
- Antenna matching circuit should be close to the antenna;
- Keep the RF traces impedance as 50Ω;
- The RF traces should be kept far away from the high frequency signals and strong interference source.

5 PCB Layout

This section will give some guidelines on PCB layout, in order to eliminate interfere or noise.

5.1 Pin Assignment

Before PCB layout, we should learn about pin assignment in order to get reasonable layout with so many external components. Following figure is the overview of pin assignment of the module.

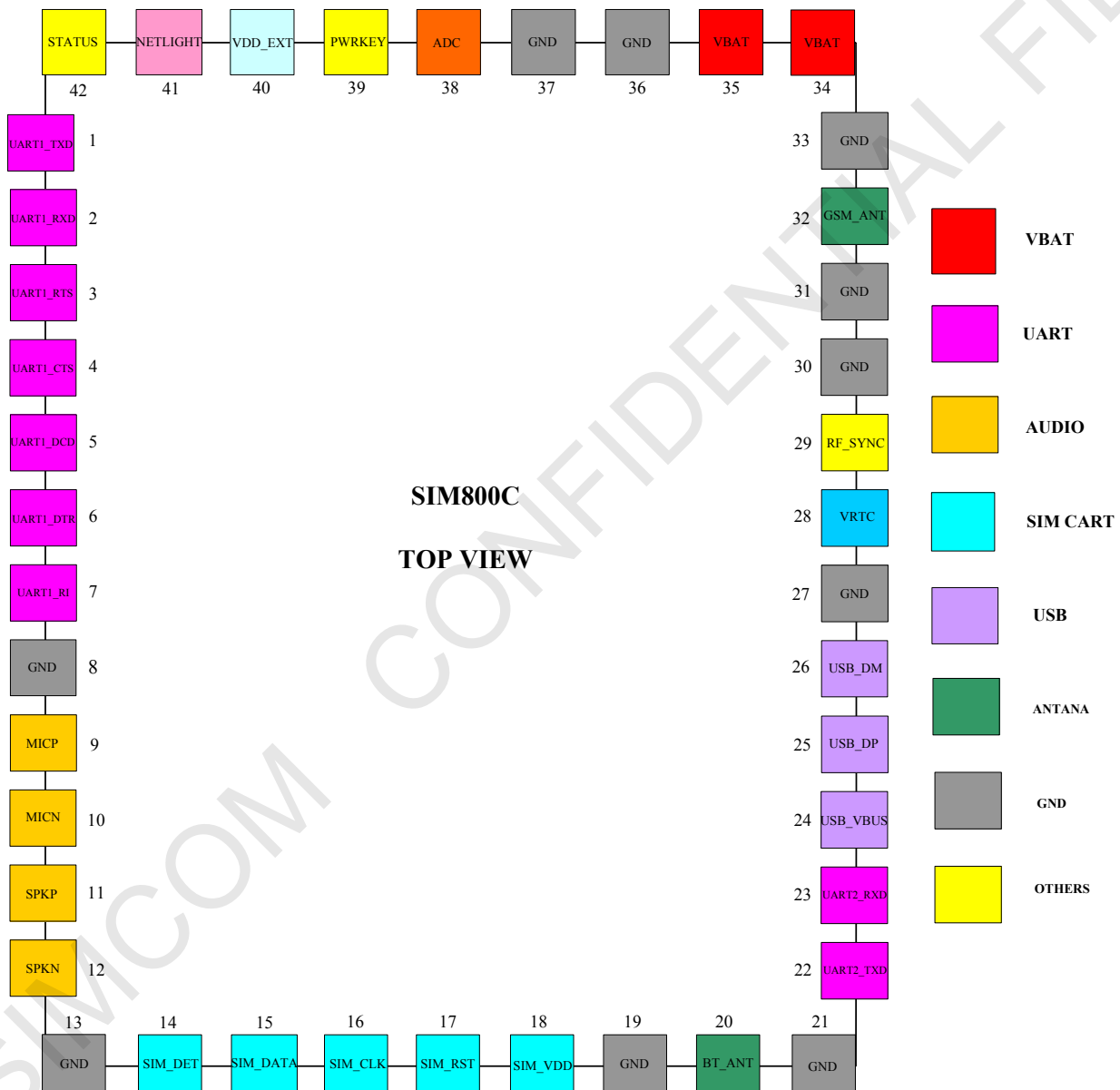


Figure 40: Pin assignment

5.2 Principle of PCB Layout

During layout, attention should be paid to the following interfaces, like Antenna, power supply, SIM card

interface, audio interface, and so on.

5.2.1 Antenna Interface

- The length of trace between pin output and connector should be as short as possible;
- Do not trace RF signal over across the board;
- The RF signal should be far away from SIM card, power ICs.

5.2.2 Power Supply

- Not only VBAT but also return GND are very important in layout;
- The positive line of VBAT should be as short and wide as possible;
- The correct flow from source to VBAT pin should go through Zener diode then huge capacitor;
- Pin 36 and Pin37 are GND signals, and shortest layout to GND of power source should be designed;
- There are 10 GND pads in the module; these pads could enhance the GND performances. On the upper layer of these pads, do not trace any signal if possible.

5.2.3 SIM Card Interface

- SIM card holder has no anti-EMI component inside. Thus SIM card interface maybe interfered, please pay more attention on this interface during layout;
- Ensure SIM card holder is far way from antenna or RF cable inside;
- Put SIM card holder near the module, as nearer as possible;
- Add ESD component to protect SIM_CLK, SIM_DATA, SIM_RST and SIM_VDD signals which should be far away from power and high-speed-frequency signal.

5.2.4 Audio Interface

- The signal trace of audio should far away from antenna and power;
- The audio signal should avoid to parallel with VBAT trace.

5.2.5 Others

- It is better to trace signal lines of UART bunched, as well as signals of USB.

6 PCB Layout

Electrical, Reliability and Radio Characteristics

6.1 Absolute Maximum Ratings

The absolute maximum ratings stated in following table are stress ratings under non-operating conditions. Stresses beyond any of these limits will cause permanent damage to SIM800C.

Table 25: Absolute maximum ratings

Symbol	Min	Typ	Max	Unit
VBAT	-	-	4.5	V
Current	0	-	2.0	A
USB_VBUS	-	-	7	V
I _I *	-	-	4	mA
I _O *	-	-	4	mA

*These parameters are for digital interface pins, GPIO, and UART.

6.2 Recommended Operating Conditions

Table 26: Recommended operating conditions

Symbol	Parameter	Min	Typ	Max	Unit
VBAT	Power supply voltage	3.4	4.0	4.4	V
T _{OPER}	Operating temperature	-40	+25	+85	°C
T _{STG}	Storage temperature	-45		+90	°C

6.3 Digital Interface Characteristics

Table 27: Digital interface characteristics

Symbol	Parameter	Min	Typ	Max	Unit
V _{IH}	High-level input voltage	2.1	-	3	V
V _{IL}	Low-level input voltage	-0.3	-	0.7	V
V _{OH}	High-level output voltage	2.4	-	-	V
V _{OL}	Low-level output voltage	-	-	0.4	V

Note: These parameters are for digital interface pins, such as keypad, GPIO and UART.

6.4 SIM Card Interface Characteristics

Table 28: SIM card interface characteristics

Symbol	Parameter	Min	Typ	Max	Unit
I _{IH}	High-level input current	-1.0	-	1.0	uA
I _{IL}	Low-level input current	-1.0	-	1.0	uA
V _{IH}	High-level input voltage	1.4	-	-	V
		2.4	-	-	V
V _{IL}	Low-level input voltage	-	-	0.27	V
		-	-	0.4	V
V _{OH}	High-level output voltage	1.62	-	-	V
		2.7	-	-	V
V _{OL}	Low-level output voltage	-	-	0.36	V
		-	-	0.4	V

6.5 SIM_VDD Characteristics

Table 29: SIM_VDD characteristics

Symbol	Parameter	Min	Typ	Max	Unit
V _O	Output voltage	-	3.0	-	V
		-	1.8	-	
I _O	Output current	-	-	10	mA

6.6 VDD_EXT Characteristics

Table 30: VDD_EXT characteristics

Symbol	Parameter	Min	Typ	Max	Unit
V _O	Output voltage	2.7	2.8	2.9	V
I _O	Output current	-	-	50	mA

6.7 VRTC Characteristics

Table 31: VRTC characteristics

Symbol	Description	Min	Typ	Max	Unit
V _{RTC-IN}	VRTC input voltage	1.2	2.8	3.0	V
I _{RTC-IN}	VRTC input current	-	3.0	5.0	uA
V _{RTC-OUT}	VRTC output voltage	-	2.8	-	V
I _{RTC-OUT}	VRTC output current	-	-	2.0	mA

6.8 Current Consumption (VBAT=4.0V)

Table 32: Current consumption

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
VBAT	Voltage			4.0		V

	Power drop	PCL=5			350	mV
	Voltage ripple	PCL=5 @ f<200kHz @ f>200kHzss			50 2.0	mV mV
I _{VBAT}	Average current	Power off mode		130	150	uA
		Sleep mode (AT+CFUN=1): (BS-PA-MFRMS=9) (BS-PA-MFRMS=5) (BS-PA-MFRMS=2)		0.9 1 1.5		mA mA mA
		Idle mode (AT+CFUN=1): GSM850 EGSM900 DCS1800 PCS1900		12.5 12.5 12.5 12.5		mA mA mA mA
		Voice call (PCL=5): GSM850 EGSM900 Voice call (PCL=0): DCS1800 PCS1900		209 211 123 143		mA mA mA mA
		Data mode GPRS (1Rx,4Tx): GSM850 EGSM900 DCS1800 PCS1900		385 389 270 285		mA mA mA mA
		Data mode GPRS (3Rx,2Tx): GSM850 EGSM900 DCS1800 PCS1900		326 340 208 228		mA mA mA mA
		Data mode GPRS (4Rx,1Tx): GSM850 EGSM900 DCS1800 PCS1900		218 228 146 162		mA mA mA mA
I _{MAX}	Peak current	During Tx burst			2.0	A

Note: In above table the current consumption value is the typical one of the module tested in laboratory. In the mass production stage, there may be differences among each individual.

6.9 Electro-Static Discharge

SIM800C is an ESD sensitive component, so attention should be paid to the procedure of handling and packaging. The ESD test results are shown in the following table.

Table 33: The ESD characteristics (Temperature: 25°C, Humidity: 45 %)

Pin name	Contact discharge	Air discharge
VBAT	±5KV	±12KV
GND	±6KV	±12KV
UART1_TXD /UART1_RXD	±4KV	±8KV
Antenna port	±5KV	±10KV
SPKP/SPKN/MICP/MICN	±4KV	±8KV
PWRKEY	±4KV	±8KV

6.10 Radio Characteristics

6.10.1 Module RF Output Power

The following table shows the module conducted output power, it is followed by the 3GPP TS 05.05 technical specification requirement.

Table 34: GSM850 and EGSM900 conducted RF output power

GSM850,EGSM900			
PCL	Nominal output power (dBm)	Tolerance (dB) for conditions	
		Normal	Extreme
5	33	±2	±2.5
6	31	±3	±4
7	29	±3	±4
8	27	±3	±4
9	25	±3	±4
10	23	±3	±4
11	21	±3	±4
12	19	±3	±4
13	17	±3	±4
14	15	±3	±4
15	13	±3	±4
16	11	±5	±6
17	9	±5	±6
18	7	±5	±6
19-31	5	±5	±6

Table 35: DCS1800 and PCS1900 conducted RF output power

DCS1800,PCS1900			
PCL	Nominal output power (dBm)	Tolerance (dB) for conditions	
		Normal	Extreme
0	30	±2	±2.5
1	28	±3	±4

2	26	± 3	± 4
3	24	± 3	± 4
4	22	± 3	± 4
5	20	± 3	± 4
6	18	± 3	± 4
7	16	± 3	± 4
8	14	± 3	± 4
9	12	± 4	± 5
10	10	± 4	± 5
11	8	± 4	± 5
12	6	± 4	± 5
13	4	± 4	± 5
14	2	± 5	± 6
15	0	± 5	± 6

6.10.2 Module RF Receive Sensitivity

The following table shows the module's conducted receiving sensitivity; it is tested under static condition.

Table 36: Conducted RF receive sensitivity

Frequency	Receive sensitivity (Typical)	Receive sensitivity(Max)
GSM850,EGSM900	< -109dBm	< -107dBm
DCS1800,PCS1900	< -109dBm	< -107dBm

6.10.3 Module Operating Frequencies

The following table shows the module's operating frequency range; it is followed by the 3GPP TS 05.05 technical specification requirement.

Table 37: Operating frequencies

Frequency	Receive	Transmit
GSM850	869 ~ 894MHz	824 ~ 849MHz
EGSM900	925 ~ 960MHz	880 ~ 915MHz
DCS1800	1805 ~ 1880MHz	1710 ~ 1785MHz
PCS1900	1930 ~ 1990MHz	1850 ~ 1910MHz

7 Manufacturing

7.1. Top and Bottom View of SIM800C

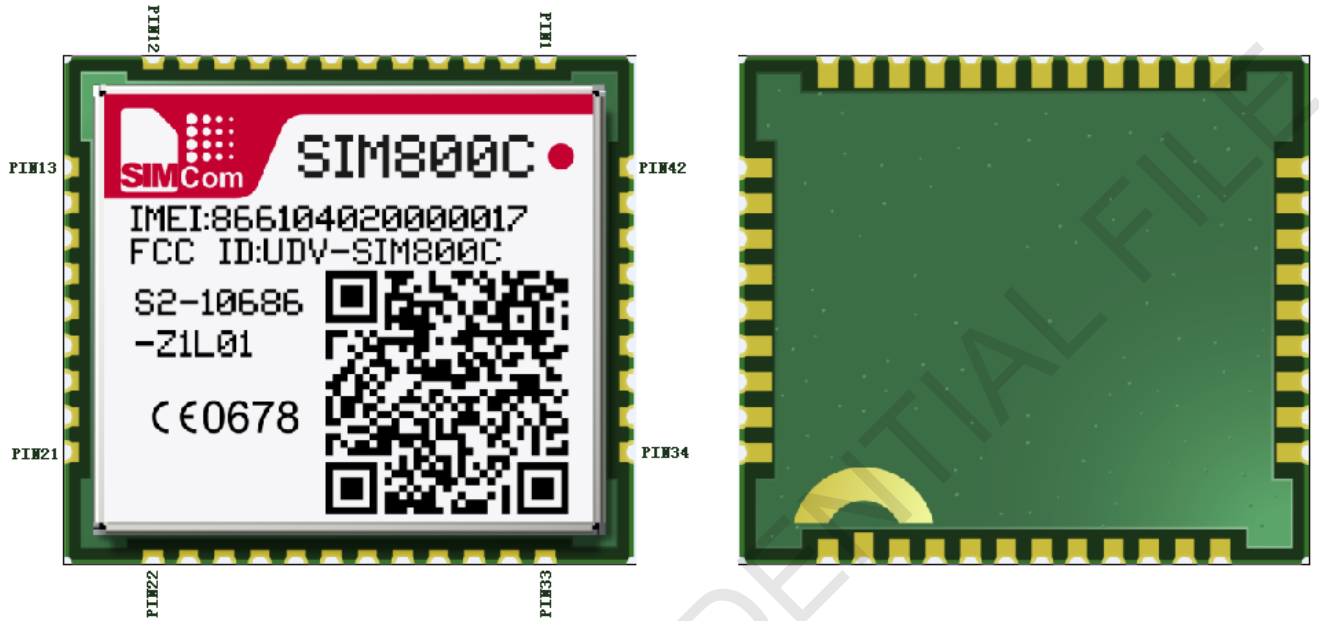


Figure 41: Top and bottom view of SIM800C

7.2. Typical Solder Reflow Profile

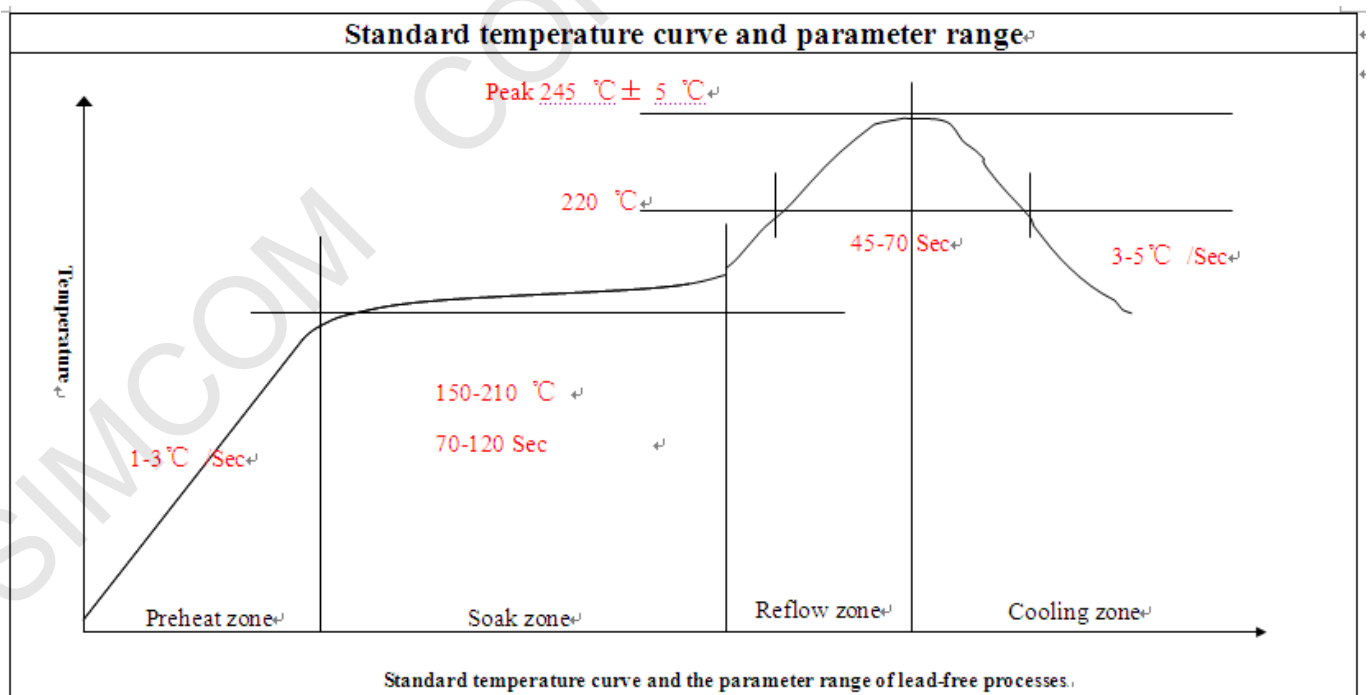


Figure 42: Typical solder reflow profile of lead-free processes

7.3. The Moisture Sensitivity Level

The moisture sensitivity level of SIM800C module is 3. The modules should be mounted within 168 hours after unpacking in the environmental conditions of temperature $<30^{\circ}\text{C}$ and relative humidity of $<60\%$ (RH). It is necessary to bake the module if the above conditions are not met:

Table 38: Moisture sensitivity level and floor life

Moisture Sensitivity Level (MSL)	Floor Life (out of bag) at factory ambient $\leq 30^{\circ}\text{C}/60\%$ RH or as stated
1	Unlimited at $\leq 30^{\circ}\text{C}/85\%$ RH
2	1 year
2a	4 weeks
3	168 hours
4	72 hours
5	48 hours
5a	24 hours
6	Mandatory bake before use. After bake, it must be reflowed within the time limit specified on the label.

NOTES:

For product handling, storage, processing, IPC / JEDEC J-STD-033 must be followed.

7.4. Baking Requirements

SIM800C modules are vacuum packaged, and guaranteed for 6 months storage without opening or leakage under the following conditions: the environment temperature is lower than 40°C , and the air humidity is less than 90%.

If the condition meets one of the following ones shown below, the modules should be baked sufficiently before re-flow soldering, and the baking condition is shown in below table; otherwise the module will be at the risk of permanent damage during re-flow soldering.

- If the vacuum package is broken or leakage;
- If the vacuum package is opened after 6 months since it's been packed;
- If the vacuum package is opened within 6 months but out of its Floor Life at factory ambient $\leq 30^{\circ}\text{C}/60\%$ RH or as stated.

Table 39: Baking requirements

Baking temperature	Moisture	Time
$40^{\circ}\text{C} \pm 5^{\circ}\text{C}$	$<5\%$	192 hours
$120^{\circ}\text{C} \pm 5^{\circ}\text{C}$	$<5\%$	6 hours

Note: Care should be taken if that plastic tray is not heat-resistant, the modules should be taken out for preheating, otherwise the tray may be damaged by high-temperature heating.

8 Packaging

SIM868 module support tray packaging.

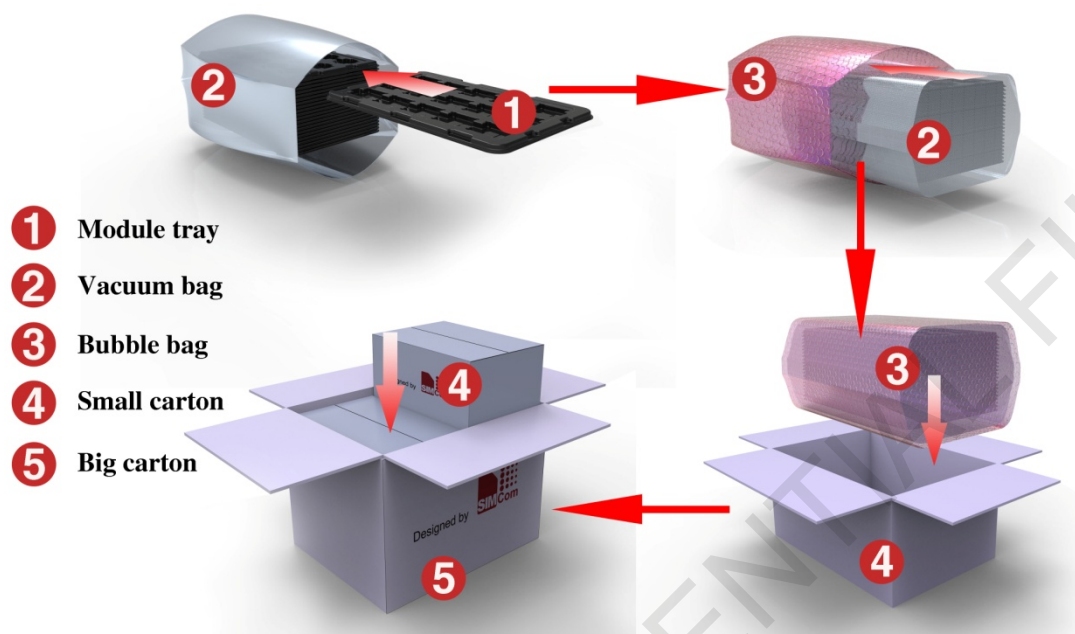


Figure 43: Packaging introduce

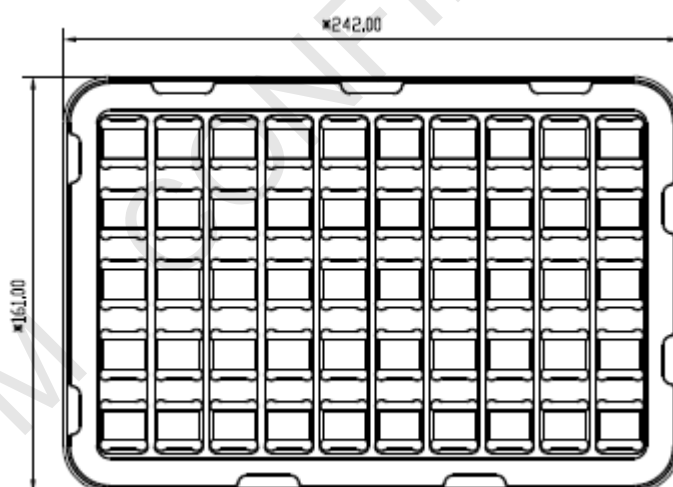


Figure 44: Module tray drawing

Table 40: Tray size

Length ($\pm 3\text{mm}$)	Width ($\pm 3\text{mm}$)	Number
242.0	161.0	50

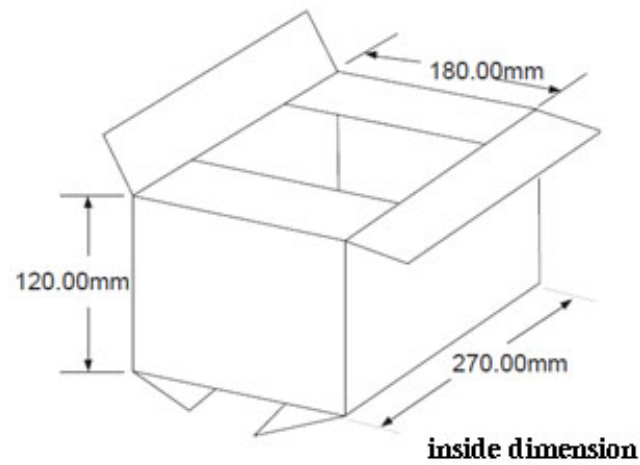


Figure 45: Small carton drawing

Table 41 : Small Carton size

Length (±10mm)	Width (±10mm)	Height (±10mm)	Number
270	180	120	50*20=1000

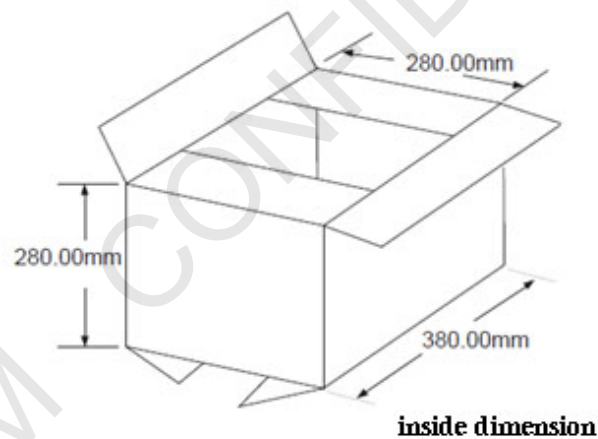


Figure 46: Big carton drawing

Table 42: Big Carton size

Length (±10mm)	Width (±10mm)	Height (±10mm)	Number
380	280	280	1000*4=4000

9 Appendix

a) Related Documents

Table 43: Related documents

SN	Document name	Remark
[1]	SIM800 Series AT Command Manual	
[2]	SIM800 Series Port Application Note_V1 02.doc	
[3]	ITU-T Draft new recommendation V.25ter:	Serial asynchronous automatic dialing and control
[4]	GSM 07.07:	Digital cellular telecommunications (Phase 2+); AT command set for GSM Mobile Equipment (ME)
[5]	GSM 07.10:	Support GSM 07.10 multiplexing protocol
[6]	GSM 07.05:	Digital cellular telecommunications (Phase 2+); Use of Data Terminal Equipment – Data Circuit terminating Equipment (DTE – DCE) interface for Short Message Service (SMS) and Cell Broadcast Service (CBS)
[7]	GSM 11.14:	Digital cellular telecommunications system (Phase 2+); Specification of the SIM Application Toolkit for the Subscriber Identity Module – Mobile Equipment (SIM – ME) interface
[8]	GSM 11.11:	Digital cellular telecommunications system (Phase 2+); Specification of the Subscriber Identity Module – Mobile Equipment (SIM – ME) interface
[9]	GSM 03.38:	Digital cellular telecommunications system (Phase 2+); Alphabets and language-specific information
[10]	GSM 11.10	Digital cellular telecommunications system (Phase 2); Mobile Station (MS) conformance specification; Part 1: Conformance specification
[11]	AN_Serial Port	AN_Serial Port

b) Terms and Abbreviations

Table 44: Terms and abbreviations

Abbreviation	Description
ADC	Analog-to-Digital Converter
AMR	Adaptive Multi-Rate
CS	Coding Scheme
CTS	Clear to Send
DTE	Data Terminal Equipment (typically computer, terminal, printer)
DTR	Data Terminal Ready
DTX	Discontinuous Transmission
EFR	Enhanced Full Rate
EGSM	Enhanced GSM
ESD	Electrostatic Discharge
ETS	European Telecommunication Standard
FR	Full Rate
GPRS	General Packet Radio Service
GSM	Global Standard for Mobile Communications
HR	Half Rate
MO	Mobile Originated
MS	Mobile Station (GSM engine), also referred to as TE
MT	Mobile Terminated
PAP	Password Authentication Protocol
PBCCH	Packet Broadcast Control Channel
PCB	Printed Circuit Board
PCL	Power Control Level
PCS	Personal Communication System, also referred to as GSM 1900
PDU	Protocol Data Unit
PPP	Point-to-point protocol
RF	Radio Frequency
RMS	Root Mean Square (value)
RX	Receive Direction
SIM	Subscriber Identification Module
SMS	Short Message Service
TE	Terminal Equipment, also referred to as DTE
TX	Transmit Direction
SINAD	Signal to Noise and Distortion Ratio
UART	Universal Asynchronous Receiver & Transmitter
URC	Unsolicited Result Code
USSD	Unstructured Supplementary Service Data
Phonebook abbreviations	

FD	SIM fix dialing phonebook
LD	SIM last dialing phonebook (list of numbers most recently dialed)
MC	Mobile Equipment list of unanswered MT calls (missed calls)
ON	SIM (or ME) own numbers (MSISDNs) list
RC	Mobile Equipment list of received calls
SM	SIM phonebook
NC	Not connect

SIMCOM CONFIDENTIAL FILE

c) Safety Caution

Table 45: Safety caution

Marks	Requirements
	When in a hospital or other health care facility, observe the restrictions about the use of mobiles. Switch the cellular terminal or mobile off, medical equipment may be sensitive to not operate normally for RF energy interference.
	Switch off the cellular terminal or mobile before boarding an aircraft. Make sure it is switched off. The operation of wireless appliances in an aircraft is forbidden to prevent interference with communication systems. Forget to think much of these instructions may lead to the flight safety or offend against local legal action, or both.
	Do not operate the cellular terminal or mobile in the presence of flammable gases or fumes. Switch off the cellular terminal when you are near petrol stations, fuel depots, chemical plants or where blasting operations are in progress. Operation of any electrical equipment in potentially explosive atmospheres can constitute a safety hazard.
	Your cellular terminal or mobile receives and transmits radio frequency energy while switched on. RF interference can occur if it is used close to TV sets, radios, computers or other electric equipment.
	Road safety comes first! Do not use a hand-held cellular terminal or mobile when driving a vehicle, unless it is securely mounted in a holder for hands free operation. Before making a call with a hand-held terminal or mobile, park the vehicle.
	GSM cellular terminals or mobiles operate over radio frequency signals and cellular networks and cannot be guaranteed to connect in all conditions, for example no mobile fee or a invalid SIM card. While you are in this condition and need emergent help, please remember using emergency calls. In order to make or receive calls, the cellular terminal or mobile must be switched on and in a service area with adequate cellular signal strength. Some networks do not allow for emergency call if certain network services or phone features are in use (e.g. lock functions, fixed dialing etc.). You may have to deactivate those features before you can make an emergency call. Also, some networks require that a valid SIM card be properly inserted in the cellular terminal or mobile.

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